Master thesis

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Update the title page with related text

Topic:

Enhancing Solar-Induced Fluorescence OCO-2 data for High-Resolution Crop Yield Prediction in Germany

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Abstract

Four sentences of background and importance of this topic.

Four sentences defining the gap analysis (work not done in field) and your problem statement.

Four sentences about the methodology of your work.

Six sentences of Results and significance of your results.

Two sentence summarizing your contribution.

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write a one page abstract - Approximately 20 sentences with importance, background, current state, gap analysis, problem statement, your solution, your methods, your results, and improvement in field being discussed.

Zusammenfassung

Da die meisten Leuten an der TU deutsch als Muttersprache haben, empfiehlt es sich, das Abstract zusätzlich auch in deutsch zu schreiben. Man kann es auch nur auf deutsch schreiben und anschließend einem Englisch-Muttersprachler zur Übersetzung geben.

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1 Introduction

For a plant, photosynthesis controls its growth, carbon uptake and effective biomass formation making its observation across large study areas important for both carbon cycle research and agricultural monitoring. Gross Primary Product (GPP) a measurable proxy for crop yield can be shown as the sum of all leaf photosynthesis [FJO+19]. Accurately measuring GPP can in turn help find crop yields at parcel, regional levels, but so far it cannot be directly measured by satellites and only remains measurable indirectly through its relationship with net ecosystem carbon exchange through ground based FLUX towers [RFB+05]. Thus only small field level data can be recorded while any regional level statistics cannot be reliably conducted. To alleviate this issue, remote sensing proxies for GPP are needed to expand the spatial scope of monitoring GPP, and one such data is Solar induced fluorescence (SIF). Solar induced fluorescence is an electromagnetic signal emitted by plant chlorophylls during photosynthesis, a higher SIF signal indicating high sunlight absorption and therefore stronger photosynthesis in stable climatological conditions. Studies like [TKM+21, MBY20, FFW+11, JYV+11] have shown that SIF has a linear relationship with GPP when working with data at satellite scales. Various studies [QLH+22, SGG+18, PGZ+20] have shown SIF to reliably estimate GPP along with vegetation stress, plant heat stress and drought like conditions at large spatial scales. Current SIF recordings have spatial sampling trade offs: OCO-2 and OCO-3 provides comparatively high resolution SIF observations along narrow and widely separated tracks while TROPOMI provides a more spatially complete coverage albeit at a much coarser resolution. Thus it is important to get a high quality SIF product to support the downstream task of GPP and in turn yield monitoring. SIF enhancement methods as shown in Table 3.1 have enhanced these satellite recorded SIF into higher resolution estimates while ensuring spatio-temporal completeness. Nevertheless, recent enhanced SIF products are limited to approximately 500m or coarser spatial resolution which is still too coarse for isolating crop pure SIF signals in fragmented European crop farms that are usually much smaller than farms in CONUS, Australia, Canada. This thesis therefore tries to address this gap by linking sparse OCO-2 SIF observations over Germany to high resolution Sentinel-2 data and secondary radiation, seasonal, and crop-composition variables represented on a 20 m grid. The resulting 20 m enhanced SIF data is treated as model derived estimates rather than as directly observed or independently validated SIF at 20 m resolution. Finally, the enhanced SIF data is used to test whether crop pure SIF signals can provide more useful information for Bavaria winter wheat yield estimation than raw spatio-temporally sparse OCO-2 footprint data containing mixed vegetation cover.

1.1 Motivation

Crop specific yield monitoring requires observations that can represent vegetation activity throughout its growing season and also cover large agricultural study regions, remote sensing data from satellites allow this. One such data Solar induced florescence (SIF) as been shown to be highly effective in monitoring crop yield and GPP [TKM+21, MBY20, FFW+11, JYV+11] FILLREF. In the preliminary work of this thesis, raw OCO-2 SIF observations were used to to investigate winter wheat yield prediction in Bavaria. Annual crop type rasters were used to filter SIF observations that had high winter wheat coverage and a seasonal SIF dataset was created for Bavaria at NUTS 3, year wise aggregation. However this procedure produced an extremely sparse dataset, with approximately 45-80% of the required values missing depending on the NUTS-3 region, year, and seasonal period and thus couldn't reliably be used to estimate winter wheat yields. This preliminary analysis demonstrated that raw OCO-2 data alone is insufficient for spatially complete and crop-specific regional yield analysis. A high resolution SIF data was needed as the average farm size in Bavaria is around 36 hectares (0.36 km^2) FILLREF and are often highly irregular in shape, higher resolution corresponds to better isolation of crop specific SIF signal.

Current raw SIF datasets have trade offs between spatial resolution, temporal coverage and measurement noise. OCO-2 provides relatively high resolution SIF observations in narrow and widely separated satellite scan tracks, while spatio-temporally complete SIF products like TROPOMI, GOME-2 have much coarser resolutions and cannot provide crop pure SIF signals at the field scale. Various enhanced SIF products have been constructed as shown in Table 3.1, but the best models are limited to 500m to 1km resolution. At these scales, one SIF pixel can overlap with multiple crops especially for small, irregular shaped farms making it difficult to match them accurately with the fixed grid cells of existing SIF products. A finer SIF representation is therefore needed to isolate winter wheat SIF signals and reduce contamination from neighbouring land / crop cover types. Out of the existing remote sensing data for vegetation indices, optical reflectance bands (prime predictors for SIF enhancement), Sentinel-2 provides the highest resolution at 10m, 20m, and 60m for the 13 reflectance bands. Combining this data with radiation, climatological, seasonal and crop composition statistics can offer a reliable way to estimate SIF at high resolution in locations and time periods without direct OCO-2 retrievals. This highlights the need to develop a more spatially and temporally complete high resolution SIF product and subsequently test whether crop pure enhanced SIF can better estimate crop yields than raw coarse resolution SIF observations.

1.2 Objective

The main objective of this thesis is to to develop and evaluate a machine learning workflow for enhancing sparse OCO-2 SIF observations combining OCO-2 supervision with high resolution spectral information as well as radiation, seasonal, and crop composition predictors. Different model architectures like tree based, multi layer perceptron (MLP) and convolutional models are compared to examine how model architecture and input data representation affects SIF estimation accuracy and the spatial distribution of the resulting output enhanced data. The study also tests whether aggregating noisy OCO-2 observations provides a more stable supervision than modelling individual footprints directly as single or multi footprint representations. The data preparation and modelling workflow is prepared in such a way that it can later be applied on a much larger spatial and temporal scope for completing high resolution SIF data. Because complete national multi year processing is computationally intensive, the thesis focuses on producing and evaluating the modelling approach, data-preparation procedures, and reproducible code rather than generating a complete Germany scale product. The proposed modelling approaches are benchmarked against relevant SIF reconstruction & downscaling methods, with capabilities and limitations highlighted FILLREF. The best model is then selected for extracting winter wheat pure seasonal SIF signals for select Bavarian NUTS 3 regions and agricultural timeframes. Finally, the study identifies whether this crop pure representation adds predictive value relative to raw coarse SIF footprints with incomplete and mixed vegetation coverage.

1.3 Thesis Scope

The main geographic scope for the thesis study is Germany, but instead of a full national coverage select Military Grid Reference System (MGRS) tiles (100km by 100km) covering the main wheat producing states of FILLREF. This spatial scope limitation is needed because multi year Sentinel-2 data processing requires substantial data storage and computational resources. Majority of the SIF enhancement in Table 3.1 use MODIS based reflectance data which can range from 250-500m resolution. In comparison, a 20m resolution raster contains approximately 156 times more pixels than a 250m raster and 625 times more than a 500m raster. Across Germany's approximately 357,600 km^2, one complete raster layer would contain about 1.43 million pixels at 500m, 5.72 million at 250m, 894 million at 20m, and 3.58 billion at 10m and this is only for one band, therefore the data storage and computation can increase drastically. The chosen 13 MGRS tiles of Sentinel-2 data is around 500GB before further model ready processing. The temporal scope is 2019-2024 because the Sentinel-2 L3A WASP monthly archive used in the thesis begins in 2019 and the required annual crop type rasters exist up to 2024. Although several secondary vegetation and radiation datasets cover longer periods, they are restricted to this shared interval to maintain a temporally consistent set of predictor channels. The thesis's SIF enhancement experiments uses available SIF observations and predictors from February to July, which is the main growing period for winter wheat in Germany. The yield estimation experiment worked with a more narrow geographic scope and covers Bavarian NUTS 3 regions within five selected MGRS tiles using monthly SIF from March to July, years 2019-2022 for model training, and 2024 as an independent temporal test year. The year 2023 and month February were excluded from the yield experiment because matching optical predictor coverage was incomplete and a complete mean aggregated SIF February season variable had a lot of missing values.

1.4 Purpose and goal

The main purpose of this thesis is to bridge the scape gap between sparse OCO-2 SIF observations and the field scale spatial information required for crop specific agricultural analysis. It aims to determine whether high resolution Sentinel-2 and secondary predictors can distribute footprint or grid supported SIF into accurate spatial patterns that are useful at the scale of fragmented German farmland. The secondary purpose is to determine if restricting enhanced SIF estimates to winter wheat pixels reduces the mixed crop cover problem encountered when raw OCO-2 SIF observations are aggregated for regional yield prediction. The study also aims to clarify how predictor information, spatial context, OCO-2 aggregation, and target preparation influences SIF enhancement performance instead of attributing all differences to model architecture alone. Beyond crop yield estimation, spatially detailed and temporally complete SIF could support the monitoring of plant phenological development, and physiological responses to drought or heat stress conditions FILLREF. But this would require changing the temporal representation of SIF, rather than representing it as daily or multi-day averages it would require sub daily SIF observations such as separate morning and afternoon estimates. Eventhough studies like FILLREF provide complete enhanced SIF products at 0.05 scale for large timeframes, this wouldn't be computationally feasible for this study given high increase in data information moving from 0.05 to 20m resolution. Instead the study contribution is a reproducible modelling and data preparation workflow that can support later generation of longer and more geographically complete SIF records. The intended outcome is therefore an evaluated high resolution SIF enhancement worflow, a transparent account of its validation limits and evidence on whether crop pure enhanced SIF provides a useful direction for future yield research.

1.5 Outline

The remainder of this thesis is organized into six chapters that progress from theoretical foundations and related research for the study, research questions, experimental worflow design, experiments conducted, their findings and conclusions.

Chapter 2 introduces the theoretical background required to understand photosynthesis and light use theory, SIF to GPP relationship, various remote sensing predictors. It also explains the modelling concepts, spatial supports and evaluation metrics.

Chapter 3 reviews existing approaches for SIF reconstruction, gap filling and downscaling methods followed by remote sensing methods for crop yield and GPP estimation. The chapter also compares main adavantages and limitations for a select set of studies and identifies the research gap addressed by this thesis.

Chapter 4 defines the direct usability problem created by the sparse spatial sampling of OCO-2 and the difficulty of obtaining crop specific SIF information. It then presents the research questions concerning SIF prediction, model representation, spatial aggregation, generalization, winter-wheat yield estimation, effect of sample density on field scale reconstructed SIF uncertainty and transferability of enhancement framework to other crops.

Chapter 5 describes the research design, study area, datasets, preprocessing procedures, and construction of the model-ready representations. It also documents the evaluated model architectures, training and validation strategies, experimental environment, reproducibility measures, and how the evaluation metrics are applied.

Chapter 6 presents and discusses the experimental results for SIF prediction, model architecture, input representation, spatial aggregation, and model generalization to previously unseen study areas. It also evaluates spatial and seasonal performance differences, compares crop pure and raw SIF for winter wheat yield prediction, and relates the findings to existing enhancement techniques.

Chapter 7 summarizes the experimental workflow and the principal findings and considers the extent to which the research questions were answered. It also discusses dissemination, practical problems and limitations encountered during the study, and possible directions for extending the SIF enhancement and crop monitoring workflow.

2 Theoretical Background

2.1 Background

2.1.1 Satellite SIF

Solar-Induced Chlorophyll Fluorescence (SIF) is a weak radiation emitted by chlorophyll cells during photosynthesis when sunlight hits the plant primary in the 650nm to 800nm wavelength. Not all of the sunlight is absorbed by the plant to avoid itself from absorbing too much energy and damaging its cells. Therefore it releases or uses the energy in three primary ways: photosynthesis fuelling plant growth, heat dissipation and SIF. SIF therefore contains information about the absorbed light and the general functioning of the photosynthetic system [DKP+22, PCTA+14]. Satellite instruments retrieve SIF from the partial filling of narrow solar Fraunhofer absorption lines in the measured radiance spectrum rather than measuring photosynthesis directly as noted in [SFJ+18, DKP+22]. For OCO-2 retrieved SIF, the wavelengths near 757nm and 771nm are captured and fitted after which the fitted fractional fluorescence contribution is converted to absolute SIF in $\mathrm{W\,m^{-2}\,sr^{-1}\,\mu m^{-1}}$ unit [DKP+22]. The retrieval quality of SIF can depend on various factors like the above mentioned spectral fit, continuum radiance, gas band ratios, solar zenith angle, cloud contamination, and the land fraction of the footprint [DKP+22]. Many of the recorded SIF observations in the OCO-2 dataset contain negative SIF values, but actual SIF cannot be negative in theory. The retrieval of a weak SIF signal combined with single sounding retrieval noise produces these negative observations with the absolute retrieval uncertainty increasing with total light reaching the satellite censor (reflected sunlight from canopy surface plus the the much weaker SIF signal) [KFM+18]. When dealing with single observations, negative SIF values must be evaluated relative to their uncertainity rather than being automagically removed. Automatic removal can increase the overall positive bias in the SIF samples, so any correction for negative samples must be done through spatial or temporal grouping of samples. For a retrieved value S_i with uncertainty σ_i, the classification was

$$q_i = \begin{cases} \text{accepted}, & S_i + 2\sigma_i \geq 0, \\ \text{questionable}, & S_i + 2\sigma_i < 0 \text{ and } S_i + 3\sigma_i \geq 0, \\ \text{rejected}, & S_i + 3\sigma_i < 0. \end{cases} \tag{2.1}$$

For this thesis, the principal SIF target combined the daily corrected retrievals in the two wavelengths as

$$S_i = \frac{S_{757,i} + 1.5 S_{771,i}}{2}, \tag{2.2}$$

with the propagated uncertainty represented as

$$\sigma_i = \frac{1}{2}\sqrt{\sigma_{757,i}^2 + (1.5\sigma_{771,i})^2}. \tag{2.3}$$

Only uncertainty based accepted observations within $-0.5 \leq S_i < 2.0$ were retained in the main datasets. The source quality flags of 0 and 1 and measurement mode nadir, glint were chosen [OPC20, OPC24]. In nadir mode, the OCO-2 satellite intrument observes directly beneath the satellite on the surface of Earth while glint mode points toward the bright reflection of sunlight on the surface. Glint observations can generally provide a stronger measured signal but have varying footprint geometries.

As noted by Joiner et al. [JYK+20] and Köhler et al. [KFM+18], the sun sensor geometry can play a major role in the absolute values of SIF. The absolute SIF values for a given region can increase greatly when the phase angle reaches 0° ie; when the sun and sensor are aligned. But if the phase angle is greater than 20°, the effect of sun geometry is small and therefore no normalization of SIF values in a given region needs to be done [KFM+18, DKP+22]. This basically means that for a given region with similar land cover and vegetation characteristics, the SIF values can vary greatly because the phase angle changes a lot between the tracks in the given region, the recorded SIF changes but the actual SIF emitted by the vegetation is almost same. Therefore it is important to correct for this geometry effect. The phase angle α for the OCO-2 SIF data was calculated from solar azimuth ϕ_s, viewing azimuth ϕ_v, SZA θ_s, and viewing zenith angle θ_v as

$$\Delta\phi = ((\phi_s - \phi_v + 180^\circ) \bmod 360^\circ) - 180^\circ, \tag{2.4}$$

$$\cos\alpha = \cos\theta_s \cos\theta_v + \sin\theta_s \sin\theta_v \cos\Delta\phi, \tag{2.5}$$

$$\alpha = \arccos(\max[-1, \min(1, \cos\alpha)]) , \tag{2.6}$$

$$\alpha_{\text{signed}} = \begin{cases} -\alpha, & \Delta\phi < 0, \\ \alpha, & \Delta\phi \geq 0. \end{cases} \tag{2.7}$$

For this thesis, phase angle and SIF value range for each recording track (single date) was compared for similar land cover and contained no such low phase angle ($< 20^\circ$) observations and showed no phase angle related increase in the SIF values, an example of this can be seen in Figure FILLREF.

2.1.2 SIF-GPP Interpretation

The relationship between SIF and vegetation productivity (GPP) can be explained through absorbed radiation data. The light use efficiency formulation from [Mon72, MM77, LLC+20] expresses gross primary production (GPP) as

$$\text{GPP} = \text{LUE} \times \text{FAPAR} \times \text{PAR} = \text{LUE} \times \text{APAR}, \qquad \text{APAR} = \text{FAPAR} \times \text{PAR}. \tag{2.8}$$

PAR is the incoming radiation in the photosynthetically active range of wavelength 400-700nm, FAPAR is the fraction of PAR absorbed by vegetation and APAR is the absorbed radiation amount by the vegetation. Light use efficiency (LUE) describes how efficiently APAR is converted into fixed carbon which is important for plant growth. At canopy level, measured SIF can be represented as [BFW13, PCTA+14, LLC+20]

$$\text{SIF} = f_{\text{esc}} \times \text{APAR} \times \Phi_F, \tag{2.9}$$

where Φ_F is fluorescence yield at the photosystem level and f_{esc} is the fraction of emitted fluorescence that escapes the canopy towards the sensor. Therefore from these formulas, SIF and GPP is generally correlated because both depend on APAR. This relationship doesn't always hold though as GPP mainly depends on LUE, whereas observed SIF depends on fluorescence yield, stress related energy partitioning, canopy structure, and escape probability as noted by Magney et al. and Dechant et al.[MBY20, DRB+20].

2.1.3 Spectral Predictors

The primary predictor variable used for enhancing SIF at high resolution was the Sentinel-2 reflectance data (composed into spectral indices). The satellite captures 13 spectral reflectance / bands ranging from visible and near infrared to short wave infrared (SWIR), with resolutions 10m, 20m and 60m. Reflectance is the fraction of incoming radiation reflected by the land surface within a defined wavelength interval. The reflectance bands chosen for this thesis are B2 blue (approximately 490nm, 10m), B4 red (665nm, 10m), B5 red edge (705nm, 20m), B8 broad near-infrared (NIR; 833nm, 10m), B8A narrow NIR (865 nm, 20 m), and B11 short-wave infrared (SWIR; 1610nm, 20m) bands [DDBC+12]. B2 provides the blue correction used by EVI spectral index, B4 corresponds to chlorophyll strength, B5 measures the transition from red absorption to high NIR reflectance, B8 and B8A respond strongly to leaf and canopy scattering, B11 respond to canopy water absorption. The five spectral indices chosen in the thesis were calculated as

$$\mathrm{NDVI} = \frac{B8 - B4}{B8 + B4}, \qquad \mathrm{NDMI} = \frac{B8 - B11}{B8 + B11}, \tag{2.10}$$

$$\mathrm{NDRE} = \frac{B8A - B5}{B8A + B5}, \qquad \mathrm{NIRv} = B8 \times \mathrm{NDVI}, \tag{2.11}$$

$$\mathrm{EVI} = 2.5 \frac{B8 - B4}{B8 + 6B4 - 7.5B2 + 1}. \tag{2.12}$$

NDVI represents the amount and condition of photosynthetically active green vegetation becoming less prominent as the vegetation canopy becomes dense [Tuc79]. NDRE represents the canopy chlorophyll content, nitrogen status in highly developed crop canopies [BCR+00]. NIRv represents the NIR reflectance directly attributable to the vegetation rather than soil and other background effects and therefore more closely related to absorbed radiation and GPP [BFB17, LLC+20]. NDMI represents the moisture content in vegetation and help understand about drought effects, it is closely related to Gao et al. NDWI measurement [Gao96]. Lastly EVI represents vegetation greennness with the atmospheric and canopy background effects reduced by the blue band and coefficient terms [HDM+02]. Given the resolution differences, the 10m bands were reduced by area averaging into 20m so that it would be compatible with the common predictor grid. The full WASP product also provides B3, B6, B7, and B12 reflectance bands. But after initial testing, these bands and their corresponding derived spectral indices were not used in the final models.

2.1.4 Spatial Reference System

All spatial datasets are provided in a specific coordinate reference system (CRS) which defines how their numerical coordinates correspond to locations on Earth. A CRS also gives additional properties such as the reference datum, map projection, coordinate units, and spatial origin. These properties allow datasets coming from different sources with different CRS to be correctly aligned and compared. Without proper CRS alignment, the numerical values can be offset by large distances and no aggregation, crop, intersection spatial operation can be conducted. There are two main CRS geographic and projected: geographic CRS represents the angular latitude and longitude while a projected CRS represents the surface in planar units (meter) and is therefore more appropriate for calculating distances, areas and fixed pixel sizes (each data value is represented by a pixel like in a image data). OCO-2 data for example is represented in native WGS 84 (EPSG:4326) CRS, it is converted to EPSG:3035 for any area, centroid calculation. The Sentinel-2 reflectance band rasters were represented in UTM CRS (EPSG:32632). To use all the predictors in a single predictor grid, all the spatial datasets must be converted to a single CRS, with the highest resolution spatial data CRS (Sentinel-2) being used as the primary CRS to avoid repeatedly reprojecting the largest raster dataset (computationally expensive). Sentinel-2 WASP data is provided in the Military Grid Reference System (MGRS) 100km by 100km tiles, it is not a CRS but a grid reference system based on the UTM CRS. It is mainly used for organizing spatial products like the Sentinel-2 product. As would be regularly mentioned in this thesis, geographic resolution in angular form like 0.5° and 0.05° do not have a fixed surface width like a 1km by 1km resolution grid cell would. For latitude φ, small angular increments $\Delta\theta$ in latitude and $\Delta\lambda$ in longitude can have its surface dimension calculated as

$$d_{\mathrm{NS}} \approx 111.2\,\Delta\theta \text{ km}, \qquad d_{\mathrm{EW}} \approx 111.2\cos(\varphi)\,\Delta\lambda \text{ km}. \tag{2.13}$$

where d_{NS} is the approximate north south surface distance and d_{EW} is the east west surface distance. For example, at approximately 50° N in Germany, a 0.05° grid cell is about 5.6km north-south by 3.6km east-west, while a 0.5° cell is about 55.6km by 35.7km.

2.2 Existing Methods

The machine learning models used for SIF enhancement can be grouped into two main groups depending on their input presentation: image based deep learning models (convolutional neural networks) and tabular data based models. The convolutional neural network (CNN) examines small groups of neighbouring pixels in an input image using convolutional filters. These filters can help learn spatial features like edges, textures, crop boundaries, vegetation patterns as we go deeper in the model's layers. The deeper layers will combine the earlier simple features to identify larger and more complex spatial patterns. For this thesis, U-Net [RFB15] a particular type of encoder-decoder CNN architecture is used. The encoder stage reduces the image size step by step while learning important features and patterns, with the deeper encoder layers using information from larger surrounding areas while the decoder enlarges the reduced image back to the original resolution while producing a prediction for each pixel ie; predicts where those learned spatial patterns are in the image. The U-Net uses backpropagation like most neural networks where a loss function first measures the difference between the prediction outputted at the end of the decoder stage with a known target, after which it calculates how much each model weight contributed to the loss error and finally the optimizer updates the convolutional filters, weights to reduce the error. This process is repeated over many training samples and epochs till the best model checkpoint is selected based on validation performance. CNN inputs for SIF enhancement are commonly represented as images with multiple stacked channels where the channels are predictor information like a coarse parent SIF value, reflectance and vegetation indices, radiation, temperature and other environmental variables. The convolution filters in the encoder stage help learn how the spatial patterns in these image channels are related to SIF. The decoder stage then converts the learned spatial patterns back to the original image resolution and produces a SIF prediction for each pixel. The training process can first coarsen the original SIF and then treat it as the reconstruction target like in Gensheimer et al. [GTK+22] or can use a transfer learning to apply a relationship learned from coarse resolution data to finer resolution predictors as shown in Du et al. [DXH+25].

A multilayer perceptron (MLP) uses fully connected layers, learned weights and nonlinear activation functions to process a set of predictor values to produce an output prediction. Unlike the CNN model, it cannot use the spatial arrangement information of neighbouring pixels in the image unless explicitly provided as a predictor as the MLP receives each input as a one dimensional list of predictors rather than as a two dimensional image. The tree based models follow a different approach: Random Forest models construct many decision trees from bootstrapped training samples and random subsets of predictors after which each tree's regression output is averaged to output a final regression [Bre01], XGBoost builds decision trees sequentially instead of independently like Random Forest, where each tree is trained to correct errors made by the pre-existing trees [CG16]. The tree based methods represent each observation as a row containing SIF and its corresponding predictor variable summaries (summarized for each SIF footprint). The models learn a scalar relationship from these rows after which the trained model is applied to spatially and temporally complete predictor grids to reconstruct SIF where observations are missing or to estimate it at a finer grid as shown in [MLC+20, CHN+22, HYD+25, CLS+25]. These tree based models are generally easier to train on irregular satellite footprints and require less memory than CNNs with the downside being unable to represent spatial, neighbourhood patterns unless explicitly feature engineered as predictors.

2.3 Evaluation Metrics

The following section will define the evaluation metrics and loss functions used for the models in this thesis. Residual is the prediction error with positive residual meaning prediction is above the observed value and negative residual meaning the opposite. The sample standard deviation of the observations s_y describes how much the observed values vary around their mean, which is needed for normalized RMSE calculation to enable reliable cross model comparison. The residual, means of observations and predictions and the sample standard deviation of the observations can be defined as

$$\begin{aligned} e_i &= \widehat{y}_i - y_i, & \overline{y} &= \frac{1}{n}\sum_{i=1}^{n} y_i, \quad \overline{\widehat{y}} = \frac{1}{n}\sum_{i=1}^{n} \widehat{y}_i, \\ s_y &= \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}\left(y_i - \overline{y}\right)^2}. \end{aligned} \tag{2.14}$$

where y_i denotes the observed value for sample i, $\widehat{y}_i$ denotes its model prediction and n is the number of evaluated samples. Pearson correlation coefficient r is the covariance between the observed and predicted values divided by the product of their standard deviations [BCHC09]. It can be defined as

$$r = \frac{\sum_{i=1}^{n}\left(y_i - \overline{y}\right)\left(\widehat{y}_i - \overline{\widehat{y}}\right)}{\sqrt{\sum_{i=1}^{n}\left(y_i - \overline{y}\right)^2}\sqrt{\sum_{i=1}^{n}\left(\widehat{y}_i - \overline{\widehat{y}}\right)^2}}. \tag{2.15}$$

The coefficient of determination R^2 is a statistical measure that shows how much an independent variable or model can explain the proportion of variation in a dependent variable. R^2 compares the squared differences between observations and its model predictions with the squared differences of each observation and the average of all observations as shown

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(y_i - \widehat{y}_i\right)^2}{\sum_{i=1}^{n}\left(y_i - \overline{y}\right)^2}. \tag{2.16}$$

A straight line is fitted through the observed and its corresponding model predicted values where the regression slope b_1 describe how much the predictions change when the observations increase and intercept b_0 is the prediction when the observed value is zero. The fitted line (regression line) can be defined as

$$\widehat{y}_i^{\text{line}} = b_0 + b_1 y_i, \tag{2.17}$$

and the slope b_1 and intercept b_0 defined as

$$b_1 = \frac{\sum_{i=1}^{n} (y_i - \overline{y}) \left(\widehat{y}_i - \overline{\widehat{y}}\right)}{\sum_{i=1}^{n} (y_i - \overline{y})^2}, \tag{2.18}$$

$$b_0 = \overline{\widehat{y}} - b_1 \overline{y}. \tag{2.19}$$

The error based metrics are root mean squared error (RMSE), mean absolute error (MAE), bias, normalized root mean squared error (NRMSE). RMSE is the square root of the mean squared residual, MAE is the mean of absolute residuals while bias is the mean of signed residuals, NRMSE is RMSE by the sample standard deviation of the observed values which describes RMSE relative to the variation of the evaluated target. These error metrics can be defined as

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (\widehat{y}_i - y_i)^2}, \tag{2.20}$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^{n} \left| \widehat{y}_i - y_i \right|, \tag{2.21}$$

$$\text{Bias} = \frac{1}{n} \sum_{i=1}^{n} (\widehat{y}_i - y_i), \tag{2.22}$$

$$\text{NRMSE} = \frac{\text{RMSE}}{s_y}. \tag{2.23}$$

Huber loss is a regression loss that, unlike RMSE, applies two different error calculations depending on the size of the residual. It uses a quadratic expression when absolute residual is within a transition value δ and a linear expression when absolute residual exceeds δ. For SIF enhancement with limited number of samples in the extreme SIF ranges, Huber loss limits the influence of a small number of large errors. For one sample, the loss can be defined as

$$\ell_\delta(e_i) = \begin{cases} \frac{1}{2} e_i^2, & |e_i| \le \delta, \\ \delta \left(|e_i| - \frac{1}{2}\delta \right), & |e_i| > \delta, \end{cases} \tag{2.24}$$

3 Related work

3.1 Satellite SIF Enhancement Techniques

Satellite observations of solar induced fluorescence (SIF) give information on vegetation photosynthesis activity but instrument characteristics in the form of information retrieval noise and uncertainties and satellite characteristics in the form of spatial footprint resolution and revisit times restrict their direct use for downstream tasks like GPP, crop yield experiments. Satellites with sensors with small footprints like OCO-2 and OCO-3 sample narrow and discontinuous ground tracks (each track is recorded in one date). Whereas wide swath instruments aboard satellites like TROPOMI [KFM+18] and GOME-2 [ABT+26] provide more complete coverage but at more coarser spatial resolution. Therefore SIF enhancement methods have been developed to address these limitations by finding relationships between native SIF and spatially complete environmental variables, relationships between SIF observations or by improving retrieval from measured radiance. SIF enhancement techniques can be broadly grouped into three categories depending on their method of enhancement. *Reconstruction and gap filling* methods estimate SIF at unobserved locations or dates and can extend a limited SIF record using an earlier predictor archive. *Spatial downscaling* methods estimate how coarse observed or reconstructed SIF is distributed among finer subpixels. A third hybrid group combines both methods and therefore contains retrieval enhancements, forecasting (prediction) and geostatistical methods for enhancement. Identifying which method was used is important when interpreting the spatial resolution of the enhanced SIF product. For example, a value reported on a 500m grid is not necessarily a SIF observation at 500m because it was inferred from learned relationship assumptions & auxiliary variables and not necessarily validated with real SIF samples at 500m resolution. Many of the methods reaggregate the fine pixel SIF estimates to the supervisory grid for validation. Therefore comparison of SIF enhancement methods considers how each method produces SIF and how it is evaluated.

3.1.1 Reconstruction and Gap Filling

Early globally contiguous SIF enhanced products confirmed that the data driven reconstruction strategies were suitable for the task. For instance, GOSIF [LX19] aggregates OCO-2 SIF at 0.05° and learns its relationship with MODIS EVI, PAR, vapour pressure deficit (VPD), and air temperature using Cubist model trees. Using the spatially and temporally complete predictor grids, they were able to complete the OCO-2 gaps at 0.05° and extend the record to year 2000. It showed that sparse OCO-2 SIF observations can support a global reconstruction. CSIF [ZJA+18] also used the MODIS satellite data specifically four reflectance bands to reconstruct instantaneous daily SIF from raw OCO-2 SIF using a simple feed forward neural network. One important finding was that differences between CSIF and observed OCO-2 SIF could reveal short term environmental stress that the predictor reflectance alone couldn't capture. Instead of applying a single model across all biomes, Yu et al. SIFoco2_005 product [YWC+19] fitted separate neural networks (ANNs) for each biome and 16 day time step, using seven-band MODIS reflectance to fill OCO-2 orbital gaps. The study found that subgrouping into separate biomes improved the distribution of training samples and the time specific models allowed the SIF reflectance relationship to vary with plant phenology and stress. Not taking care of this biome and time stratification caused seasonal under-prediction and over-prediction and reduced drought sensitivity of the model. SIFoco2_005 product was externally tested with air plane recorded SIF values (using the Chlorophyll Fluorescence Imaging Spectrometer (CFIS)) and thus provided a much stronger validation than same sensor testing. Though like the previous studies the resulting output was only viable at 0.05°, much coarser resolution than what would be needed for crop pure field scale studies. The spatially continuous TanSat product by Ma et al. [MLC+20] used a random forest model, MODIS optical variables and cosine of solar zenith angle for SIF gap filling. This study showed that cosine of solar zenith angle was a strong proxy for the amount and geometry of incoming sunlight (illumination), since SIF intensity depends partly on sunlight absorbed by vegetation this variable was important for SIF enhancement.

The previous OCO-2 reconstructions mainly focused on spatial completion of the spatially sparse OCO-2 records, but reconstruction can also extend a temporally short satellite record over a longer historical period. TROPOMI provides near-global daily coverage of SIF at 0.05°, but its observations only became available in 2018 and still contain gaps caused by clouds and quality filtering, therefore temporal completion could be applied for this dataset. Chen et al. [CHN+22] addresses this with their RTSIF product which used an XGBoost model to train on clear sky TROPOMI SIF with predictors; MODIS reflectance, land-surface temperature, land cover, PAR, C3/C4 vegetation fractions after which the trained model was applied to historical predictor dataset to extend the range of TROPOMI SIF historically. The variables chosen for modelling are important in the light use efficiency model FILLREF, where reflectance characterizes canopy state and absorbed radiation, while temperature and vegetation type help represent changes in fluorescence efficiency. Compared to the previous studies, DOSIF pursues a more finer temporal objective by producing daily, globally contiguous OCO-3-like SIF [YZW+25]. It used a moving spatial-temporal window sampling strategy where a separate model was constructed for each of the year using observations from nearby dates, multiple years, and geographically distinct sub-biomes. CatBoost model was found to be the best model for DOSIF's strategy. Again like [YWC+19], this sampling strategy outperformed a single global model indicating that relationships between SIF and reflectance vary across regions and stages of vegetation development (seasonal growth of vegetation).

ST-LGBM [SWG+22] like the previous OCO-2 SIF supervisory source studies tried to address the demanding problem of SIF gap filling large spatial gaps between OCO-2 track records. They used NIRv, PAR, temperature, vapour-pressure deficit, and land cover data in tabular format and therefore couldn't naturally contain neighbouring spatial information unlike in an image format. They addressed this by making spatial SIF factors from geographically nearby observations with similar vegetation state and temporal factors from observations at similar stages of the vegetation growth cycle in other years. This allowed the LightGBM model to use information from neighbouring regions (swaths) rather than relying only on variables observed at the same location and time without neighbourhood information. When they tested the LightGBM model without the feature engineered spatial and temporal factors on the OCO-2 SIF incomplete regions it performed much worse than the model with the spatial and temporal factors. This demonstrated that random validation may overestimate performance in areas without OCO-2 observation unless provided with the spatial and temporal factors. An even more finer temporal reconstruction can also be done as demonstrated by He et al. [HYD+25]. Their study fit separate XGBoost models with OCO-3 SIF near 10:00 and 14:00 local time and GOCI reflectance bands as predictor variables producing continuous 500m resolution SIF maps for the morning and afternoon satellite overpasses. They normalized SIF using PAR and NIRv to produce fluorescence yield to reduce the influence of incoming sunlight and canopy structure and used the morning-to-afternoon normalized SIF differences as a drought indicator. The study showed that water and heat stress can be recorded in sub-daily observation (reduction in photosynthetic activity occurring within the same day, plants can close their stomata early during a heat stress), such effect cannot be realised in a single SIF day observation. These reconstruction studies showed that SIF reconstruction methods have developed from simple models applied globally to models that consider location, time, nearby SIF observations and changes within a day. They also show that model performance depends on how the validation was done and whether the input variables capture short term changes in plant activity rather than only capturing physical properties of the vegetation canopy (optical variables like NDVI, NIRv etc.).

3.1.2 Spatial Downscaling

Spatial downscaling means that a relationship learned between a coarse target variable (SIF) and a set of predictors can be evaluated using the same predictors at a high resolution to estimate the target variable at a finer resolution. Zhang et al. [ZLQ+24] used GOSIF [LX19] as the target trained with a random forest with predictors; NDVI, daytime land surface temperature at 0.05° resolution after which the trained model was applied on 1km predictors to produce enhanced GOSIF at 1 km resolution. The enhanced GOSIF supports a stronger association with anomaly drought index and MODIS GPP than the original GOSIF. Similarly a related study instead uses a CNN model with NDVI, EVI, day and night time temperature to downscale GOSIF to approximately 1 km [ZXQL20]. Unlike the tabular data representation, the convolutional representation incorporates adjacent pixel context which was found to be useful for model fitting. Thought it should be noted model evaluation mainly relied on reaggregated (to match their spatial resolution)enhanced GOSIF with original GOSIF and relationships with GPP, drought, or yield. These results as with almost all study results demonstrate only an application value and does not independently validate the enhanced SIF at the fine scale, as the best instrument recorded SIF resolution is around 1km^2 - 2km^2 (OCO-2, resolution varies with latitude). Also since GOSIF itself is an enhanced SIF product, its errors will carry over to any further enhancement experiments. Duveiller et al. downscale GOME-2 SIF using finer resolution NIRv, NDWI, and daytime land surface temperature [DFW+20]. External validation with OCO-2 is used to choose the best model configuration. It was also successfully validated with TROPOMI after a small pixel level correction. Unlike methods mentioned earlier that tries to predict SIF independently, this approach distributes the measured GOME-2 SIF within each coarse pixel to a finer resolution.

Hu et al. [HJM+24] BCSIF product first used a random forest model to predict high resolution TROPOMI SIF from spectral indices, illumination and environmental variables after which they estimated the difference between observed coarse resolution TROPOMI and predicted aggregated SIF and then redistributed this bias to a 0.005° grid. This bias correction helps preserve the physiological information present in TROPOMI base SIF but not captured by the relationship between SIF and its predictors. Jin et al. [JGF+24] used random forest to predict high resolution SIF from coarse GOME SIF and predictors from AVHRR and ERA5. Residuals are then produced from the aggregated high resolution predictions versus the coarse GOME SIF. They then use ordinary kriging to estimate residuals in locations without observations based on residuals of nearby regions, after which the estimated residual is then added to the earlier high resolution random forest prediction thereby correcting its local errors. Their use of predictors from AVHRR and ERA5 helped SIF reconstruction before the MODIS era. SIFnet [GTK+22] first coarsened TROPOMI SIF to 0.5° and then trained a CNN (configured to learn the residuals) to recover the original 0.05° SIF map from the coarsened SIF (0.5°) and 0.05° resolution secondary predictor variables (upscaled from original 0.005°), thereby learning a tenfold scale transformation. They assumed that this scale transformation can then be applied to 0.05° native TROPOMI and 0.005° secondary predictors to produce a 500m resolution SIF map. They validated the 500m resolution downscaled SIF by aggregating to OCO-2 and OCO-3 footprints, reporting RMSE = 0.21, R^2 = 0.57 and RMSE = 0.19, R^2 = 0.62 respectively. The parent coarsened TROPOMI SIF was the important feature followed by NIRv, Cosine of the solar zenith.

TroDSIF [CLS+25] like the previous tabular methods uses random forest to predict SIF at 500m resolution, but instead of treating it as a final SIF map, the predictions are then used to determine how the original coarse TROPOMI SIF should be distributed among the smaller pixels by means of Gaussian redistribution. Therefore, the high resolution SIF map preserves the original TROPOMI measurement while adding spatial detail assuming that there was strong linear proportionaility between the original SIF and predicted SIF (only then redistribution is valid). CNSIF combines long term SIF reconstruction with CNN based downscaling [DXH+25]. The CNN uses GOSIF, NIRv, EVI, near-infrared reflectance, and land-surface temperature after which transfer learning is used to produce 500m estimates from Landsat data. It was validated on held out 2019 GOSIF samples rather than OCO-2 or OCO-3 SIF samples. Like the previous GOSIF enhanced studies, CNSIF will also inherit the uncertainities from GOSIF SIF estimates. HSIF [ZZZ23] instead of using a statistical learning method like the previous studies uses a physical based light use efficiency and radiative transfer method to determine how coarse TROPOMI SIF should be distributed to 1 km resolution. The relationships are determined using physical variables and equations, such as absorbed radiation, fluorescence efficiency, and canopy escape probability, rather than by fitting a machine learning model to labelled SIF samples. Overall the downscaling approaches show that high resolution predictor variables can add useful spatial detail to coarse SIF observations. But the reliability of this detail depends on whether the downscaling method preserves the original SIF measurement and if the downscaled SIF product was validated using high resolution original SIF measurements like from OCO-2 or OCO-3.

3.1.3 Other and Hybrid Enhancement Approaches

Some SIF enhancement studies enhance SIF using hybrid approaches that use a mix of various techniques like reconstruction, gap filling, downscaling and geostatistical approaches. Pais et al. [PBM25] firstly gap fills in the sparse OCO-2 SIF dataset with a random forest model using MODIS spectral indices, reflectance bands and land surface temperature variables as predictors, after which it then forecasts monthly and seasonal SIF using two earlier SIF maps, EVI, spatial and temporal variables. They compared various statistical and deep learning models, but an important point is errors from the first gap filling step can affect later forecasts. Li et al. [LCD+25] instead tries to denoise the original SIF retrievals by averaging matched OCO-2/3 observations within TROPOMI footprints and using them to train ANNs on TROPOMI light measurements at different wavelengths. This approach reduces noise in individual SIF retrievals and can therefore provide a better training data for later reconstruction or downscaling. Tadić et al. [TII+24] uses the previous CSIF enhanced SIF dataset to get an initial SIF estimate at every location, after which they then compared CSIF values with nearby OCO-2 SIF measurements. They then used kriging (geostatistical approach) to transfer differences between the CSIF and OCO-2 values to nearby local CSIF areas that have no OCO-2 measurements, thereby using the difference to improve the estimate. coSIF [JCZM23] uses the cokriging method to model monthly OCO-2 SIF with atmospheric carbon dioxide from the following month. This method essentially learns how SIF values are related across the spatial setting and also how SIF is related to later changes in carbon dioxide. With this learned relationship, it can then use nearby SIF and carbon dioxide data to estimate missing SIF values.

In conclusion, there is no single SIF enhancement strategy that can resolve all the limitations of satellite acquired SIF data. Global reconstructions of SIF can provide continuous SIF enhanced records but may smooth biome specific conditions, studies have shown making separate models for each biome can improve the SIF enhancement performance. Non observation preservation downscaling approaches likewise can also produce high resolution SIF estimates but most of these methods use validation between these aggregated enhanced SIF maps with the coarse SIF value and strong relationship with GPP. These validation techniques cannot guarantee if the high resolution SIF map is accurate at the spatial support level because incorrect high resolution values can average to the correct coarse value, ie; the spatial distribution of the fine pixels in the coarse SIF can be wrong. Unlike the previous method, observation preserving and physical approaches can stay true to the original coarse SIF signal (ie; they average to original value) but again spatial aggregation of the fine pixels can still be inaccurate. Geostatistical methods (studies that used kriging method on OCO-2 SIF data) take care of local observation structure and SIF retrieval uncertainty. Though one critical issue with these methods is that their SIF gap-filling performance weakens as gaps between original SIF samples grow and nearby observations disappear. Using the OCO-2 dataset which is highly sparse spatially and temporally can greatly decrease the viability of these methods on a lot of regions. All of these methods have their advantages and disadvantages, the usage depends on the availability of the respective predictor data, quality of the original SIF signals in the area of study.

Table 3.1: Summary of SIF enhancement methods for reconstruction, spatial downscaling, forecasting, retrieval improvement, and geostatistical gap filling. The † indicates an unofficial acronym.

Product	**SIF source**	**Predictors**	**Model family**	**Spatio-Temporal resolution**	**Period**	**Internal validation metrics**	**External validation metrics**
Reconstruction (no aggregation constraint)							
GOSIF [LX19]	OCO-2	MODIS EVI; PAR, VPD, air temperature	Cubist model trees	Global; 0.05°; 8-day	2000-2017	$R^2 = 0.79$; RMSE = 0.07	FLUXNET GPP proxy: $R^2 = 0.73$
TanSIF [MLC+20]	TanSat	MODIS reflectance, NDVI, cos(SZA), air temperature	Random forest	Global; 0.05°; 4-day	2017-2019	$R^2 = 0.73$-0.81; RMSE = 0.30-0.32	TROPOMI SIF: $R^2 = 0.73$
DOSIF [YZW+25]	OCO-3	MODIS 7-band reflectance; biome and temporal context	MSTWS + CatBoost	Global; 0.05°; daily	2001-2024	$R^2 = 0.81$-0.85; RMSE = 0.07-0.08	Airborne CFIS: $R^2 = 0.75$; RMSE = 0.11
RTSIF [CHN+22]	TROPOMI	MODIS reflectance, LST and land cover; PAR; C3/C4 fractions	XGBoost	Global; 0.05°; 8-day	2001-2020	$R^2 = 0.907$; RMSE = 0.062	PhotoSpec tower SIF: $R^2 = 0.754$-0.84
ROSIF [HYD+25]	OCO-3	Eight GOCI reflectance bands	XGBoost	Liaoning; 500 m; 10:00/14:00	2019-2020	$R^2 = 0.598$-0.684; RMSE = 0.348-0.388	Drought proxies: SPEI $r = 0.71$; SMZ $r = 0.53$
ST-LGBM [SWG+22]	OCO-2	NIRv, PAR, VPD, temperature, land cover; spatial/temporal SIF factors	Constrained LightGBM	Global; 0.05°; 8-day	2014-2019	$R^2 = 0.79$-0.83; RMSE = 0.07	TROPOMI SIF: annual $R^2 = 0.91$
SIFoco2_005 [YWC+19]	OCO-2	MODIS 7-band reflectance; biome/time stratification	Feed-forward ANN	Global; 0.05°; 16-day	2014-2018	$R^2 = 0.83$; RMSE = 0.065	Airborne CFIS outside swaths: $R^2 = 0.72$; slope = 0.96
CSIF [ZJA+18]	OCO-2	MODIS red, NIR, blue and green reflectance	Feed-forward NN	Global; 0.05°; 4-day	2000-2017	$R^2 = 0.786$; RMSE = 0.177	FLUXNET GPP proxy: median $R^2 = 0.64$; RMSE = 1.67
Spatial downscaling							
RF-GOSIF† [ZLQ+24]	GOSIF	MODIS NDVI and daytime LST	Random forest	Henan; 1 km; monthly	2001-2020	Reaggregated GOSIF: $r = 0.74$	MODIS GPP proxy: $r = 0.74$; crop yield: $r = 0.89$-0.93
G2DSIF† [DFW+20]	GOME-2	NIRv, NDWI and afternoon LST	Semi-empirical LUE redistribution	Global; 0.05°; 8-day	2007-2018	-	OCO-2 SIF: $r = 0.813$-0.825; abs. bias = 0.0615-0.0630

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Table 3.1 - continued

Product	**SIF source**	**Predictors**	**Model family**	**Spatial / temporal resolution**	**Period**	**Internal validation metrics**	**External validation metrics and data**
BCSIF [HJM+24]	TROPOMI	MODIS reflectance, NDWI, cos(SZA); temperature, VPD, soil moisture	RF + bias correction	China; 0.005°; 16-day	2019-2020	$R^2 = 0.887$; RMSE = 0.0716	ChinaSpec tower SIF: $R^2 = 0.798$; RMSE = 0.141
CNN-GOSIF† [ZXQL20]	GOSIF	NDVI, EVI, daytime and nighttime LST	2D CNN	Henan; 0.008°; monthly	2013-2017	Reaggregated GOSIF: $r \leq 0.9174$	MODIS GPP proxy: $r \leq 0.8346$; Maize yield: $r = -0.84$
DSIFRFK [JGF+24]	GOME	AVHRR NIRv, NDVI and NIR; temperature, radiation, precipitation	RF + residual kriging	East Asia; 0.05°; monthly	1995-2003	$R^2 = 0.83$; RMSE = 0.08	Other SIF products: r = 0.73-0.89; tower GPP: R^2 = 0.61-0.99
SIFnet [GTK+22]	TROPOMI	Coarse SIF; MODIS reflectance/VIs; climate, soil, land cover, elevation	Residual CNN	CONUS; 0.005°; 16-day	2018-2021	$R^2 = 0.92$; RMSE = 0.17	OCO-2/3 footprints: R^2 : 0.57-0.62; RMSE : 0.19-0.21
TroDSIF [CLS+25]	TROPOMI	MODIS reflectance, NDVI, cos(SZA), air temperature	RF + Gaussian redistribution	Global; 500 m; 16-day	2018-2021	Validation $R^2 = 0.8935$; RMSE = 0.064; reaggregation R^2 = 0.934-0.948	Six ChinaSpec sites: RMSE = 0.104-0.223; MAE = 0.077-0.163
HSIF [ZZZ23]	TROPOMI	Chlorophyll fPAR, fluorescence efficiency, escape probability, PAR, land cover	Physical LUE/radiative-transfer model	Global; 1 km; source-day/8-day	2018-2020	-	OCO-2 total SIF: $R^2 = 0.78$; RMSE = 1.33; tower GPP proxy: $R^2 = 0.70$
CNSIF [DXH+25]	GOSIF	GOSIF, NIRv, EVI, NIR and LST from Landsat	2D CNN + transfer learning	China; 500 m; monthly	2003-2022	$R^2 = 0.8566$; RMSE = 0.089	Nine ChinaSpec sites: pooled $R^2 = 0.581$; RMSE = 0.152; site R^2 = 0.324-0.947
Other and hybrid enhancement approaches							
OCO2-FSIF† [PBM25]	OCO-2	Two lagged SIF fields, EVI, spatial and temporal variables	RF gap fill; Lasso/LightGBM forecast	India; ∼1 km; monthly/seasonal	2017-2019	RMSE = 0.0355-0.0594; MAPE = 16.91-19.22%	-
ANN-TroSIF† [LCD+25]	Matched OCO-2/3	TROPOMI radiance spectra at 743-758 nm	Column-specific feed-forward ANNs	Global; native TROPOMI footprint; daily	2018-2024	$R^2 = 0.8535$; RMSE = 0.217	Flux-tower GPP proxy: R^2 = 0.45-0.79
KED-SIF† [TII+24]	OCO-2	CSIF external drift and covariance of nearby OCO-2 SIF	Kriging with external drift	Global; 0.05°; 4-day	2019	$R^2 = 0.8523$; RMSE = 0.1556	-
coSIF [JCZM23]	OCO-2	SIF/XCO_2 covariance and spatial basis functions	Multivariate cokriging	North America; 0.05°; monthly	2021	Spatial blocks: RASPE = 0.56-0.58; bias = −0.06-0.01	-

3.2 Remote Sensing-Based Crop Yield and GPP Estimation

A crop's yield can be dependent on various factors ranging from photosynthetic carbon uptake, respiration ratio, biomass allocation, effects from sudden shifts in weather conditions, soil and biological & heat stress all of which is factor during its growing season FILLREF. Remote sensing from satellites can only capture some of these factors from optical radiometers and spectrometers. The signals that they can capture they do so usually in high temporal and spatial coverage. The following reviewed studies can be grouped in four approaches: SIF based yield estimation, optical and spectral index, multi source yield estimation and finally SIF based GPP estimation. Some of these groups can overlap in terms of scope as they first estimate the GPP and then derive the yield from its direct relationship FILLREF while other studies use both SIF and spectral index predictors for estimation. The main SIF based yield estimation studies directly related reported yield (target) with seasonal (month wise) SIF using machine learning models, thereby showing data driven statistical approaches were suitable for estimating crop yields [PGZ+20] [ZHN+25] [JPCB23]. Peng et al. [PGZ+20] compared the yield predictive power of various SIF products: gap filled OCO-2, TROPOMI and much coarser GOME-2 SIF along with secondary variables like MODIS indices, land surface temperature for maize and soybean in the U.S Midwest. They showed that the higher resolution gap filled OCO-2 and TROPOMI

SIF data outperformed GOME-2 by a significant margin and thus showed that the higher resolution SIF signals contain more crop pure SIF signals. They also noted NIRv (spectral index) can sometimes match SIF in performance depending on the crop type, validation method and training record length, though SIF still remained the dominant predictor variable for majority of the tests. Their tests included select regions in the North China Plain and CONUS (US), where a SIF based XGBoost reported a $R^2 = 0.87$ for the former while the CONUS study found that adding SIF to NIRv explained yield variation from 64% to 69%.

Studies such as [QLH+22], [SGG+18] found that SIF is particularly informative when stress induced from drought, heatwave affects the photosynthesis process well before any visible canpoy structure changes. As a reminder SIF is a faint light signal emitted by the plant's chlorophyll when absorbing sunlight during photosynthesis, while spectral indices like NIRv, NDVI, EVI measure plant canopy structure, greenness, the stress can affect the plant photosynthesis process before any changes can be seen in canopy. [QLH+22] found that during the 2012 U.S Midwest drought yield declined by 25%, where predictor variables GOSIF (enhanced SIF product) declined by 22% ($R^2 = 0.91$) while NDVI, EVI and NIRv declined by only 4-10%. Likewise [SGG+18] found that SIF detected reduced winter wheat growth activity in March well before NDVI and EVI showed any changes for the 2010 Indian heatwave. They also showed that SIF had a bigger response to yield than fPAR (both the variables are connected by the light use efficiency model FILLREF), thereby indicating that the plant reduced its photosynthetic efficiency rather than only absorbing less sunlight. Though it should be noted that early stress detection does not guarantee accurate final yield estimation as SIF predicted a 13.9% yield loss compared with the reported yield loss of 6%.

Unlike the previous studies, yield estimation studies like [GBZ+16] [HMD+20] [KWH+24] [XHL+25] [WYL+25] first estimate Gross Primary Product (GPP) from SIF and then estimate crop yield from GPP after some corrections (aslo called process guided models). The common predictors used for these models include corrected canopy escape probability, C_3 and C_4 composition, respiration, NPP:GPP ratio, harvest index and harvested area. He et al. [HMD+20] found that seasonal TROPOMI SIF successfully validated with ground based flux tower (eddy covariance tower) GPP measurements with a $R^2 = 0.89$, with the secondary predictor C_3/C_4 composition increasing productivity relationship from $R^2 = 0.72$ to 0.86. Kira et al. [KWH+24] showed that a mechanistic light reaction model (non statistical based) was comparable to machine learning based techniques like ANN, random forest for U.S. maize yields. For regions like the Indo-Gangetic plain that had limited high quality remote sensing data, the mechanistic model in fact outperformed the machine learning models showing alternative ways of estimating GPP and crop yields. Another way of estimating yield was using q_L (proportion of PSII reaction centres that are open. The centres help convert light into chemical energy. A higher q_L thereby means more capacity for photosynthesis) and SIF to estimate plant photosynthesis. Wang et al. [WYL+25] used this method to explain about 78% of differences in county level maize and soybean yields $R^2 = 0.78$. These mechanistic models are more interpretable than regression models as they use the direct physical process for yield estimation but they still depend on crop specific parameters and assumptions that are used to convert photosynthetic uptake to yield. Therefore the choice between direct SIF-yield methods and SIF-GPP-yield methods depends on data availability. The machine learning models can learn complex relationships when there are long, reliable yield records whereas the process guided models need fewer yield records (but require more plant specific data like respiration, crop allocation, q_L).

Older spectral index only approaches like [VG21] [CTS+19] [HBC+19] [VHI+22] tried to estimate crop yield from canopy greenness and structure, chlorophyll, plant water content (each variable has a related spectral index). Seasonal NDVI summaries were found to be weak winter wheat yield predictors compared to precipitation data observed during wheat tillering and flowering stage ($R^2 = 0.66$) for Belgium winter wheat yields [VG21]. In contrast [CTS+19] found that NDVI and NDWI (recorded during heading stage; the stage when the grain head first emerges) explained roughly 95-96% of grain yield variation on a small set of farms in central India, but it was not validated on regional scale. Hunt et al. [HBC+19] used Sentinel-2 spectral indices and showed a yield prediction performance of RMSE from 0.66 to 0.61 t ha^{-1} on U.K. wheat yield data. Vallentin et al. [VHI+22] conducted a yield estimation study for Northeast Germany (13 years) and showed that spectral indices correlated with yield ranging from almost 0 to 0.94, with a median of only 0.19, thereby showing an overall weak correlation between spectral indices and yield. They also found that the correct data acquisition timing, field heterogeneity (high resolution spectral indices at 10m can capture spatial details in farms that were heterogeneous) mattered more for high spectral index-yield correlation than choice of a particular spectral index. Overall the spectral index only approaches showed a relatively weaker yield estimation performance than the main SIF based approaches.

Multi source studies combine spectral index data (which describe plant canopy characteristics), weather and soil data (growth constraints) and crop-growth simulations (which provides data for water stress) to estimate yields as shown in studies like [Goi24] [MKS22] [BBB+24] [ZPZ+20] [SSK+22]. In southern Germany, raw Sentinel-2 bands along with precipitation and evapotranspiration data explained 84% of winter wheat yield variance [MKS22]. Likewise [ZPZ+20] showed for a northeastern Australian set, Sentinel-2 index predictors along with crop-growth simulation water stress index and chlorophyll indices could explain 93% of wheat yield differences. Brandt et al. used six multi source estimators to conduct yield estimation at the parcel (crop field) level with roughly 140,000-155,000 parcel samples available per year until 2019. The reported a $R^2 = 0.74$ at parcel level and 0.79-0.89 at the district level (NUTS 3) after aggregation. Yield performance from these studies cannot be directly compared because their target resolution ranges from parcel (field) level to county or district averages. Some of the studies tested on a random test sample which could contain same year and region samples and therefore cannot test genuine extrapolation from training to test. Studies that tested on aggregated county or district samples can report a much higher R^2 as aggregation reduces local noise.

The remaining yield studies like [HM20], [MGW+25], [SBG22] focused only on estimating GPP and not the final crop yield from SIF, but it still provides a physiological link between SIF and biomass / yield production. Hu et al. [HM20] found that using local relationships to down scale GOME-2 SIF represented regional and seasonal changes in GPP better than using one global relationship to downscale GOME-2 SIF. Ma et al. [MGW+25] used a transfer learning approach to combine the strengths of two datasets: SIF-derived GPP and Eddy-covariance (ground based FLUX tower) GPP. Their transfer learning model first trained on the SIF derived GPP data to learn general relationships between GPP and predictors like NIRv, leaf area index, incoming solar radiation, temperature and vapour pressure deficit after which the pretrained model was then fine tuned using GPP measurements from 164 FLUX tower sites. This gave an improvement of $R^2 = 0.773$ compared with the pretrained model showing $R^2 = 0.641$. Though such a study is limited in geographic scope and is highly local, transferability to other regions depends on availability of nearby FLUX tower data. It should also be noted that an accurate GPP estimation does not necessarily determine the crop yield. Other factors like respiration, timing of carbon uptake, allocation to non grain biomass, harvest index affect the final conversion of GPP into yield. These studies and the earlier main SIF based approaches confirm that SIF is the important variable for crop yield / GPP estimation as it can directly measure photosynthetic activity of the plant. It is also useful for detecting heat stress in crops and a good proxy for identifying droughts conditions. Other auxiliary data like spectral indices, weather, soil, crop composition can add useful information with SIF to better explain the yield process. Common limitations of these methods are low resolution satellite records (mainly for SIF data) giving poor local / parcel level information, temporal and spatial gaps in the data, uncertain target crop area and yield statistics which can affect the modelling process, and spatial aggregation of yield statistics causing accuracy gains / noise reduction compared to local samples. These findings therefore support that spatially enhanced SIF along with high resolution spectral and environmental variables are useful for crop yield or GPP estimations.

3.3 Comparative Analysis and Research Gap

This section briefly identifies important findings and methodological gaps in existing SIF enhancement methods that can be addressed by future studies. Spatially and temporally continuous SIF was shown that it can be reconstructed from sparse satellite observations using reflectance, vegetation indices, radiation, temperature, and other environmental variables using either statistical data driven approaches or mechanistic models depending on the availability of data. Most of the discussed methods that preserve the original coarse SIF observation as part of their learning process provide more consistent results than completely unconstrained prediction, though this can be challenging with sparse OCO-2 and OCO-3 datasets. Discrepancy between output enhanced SIF resolution and available SIF products at high resolution for validation is a challenge that needs to be addressed. Some studies have used air plane [YWC+19], [YZW+25] or FLUX tower based SIF recordings [HJM+24], [CLS+25], [DXH+25] (high resolution) to validate their enhanced SIF product, but the validation remains viable only at the local level and cannot be generalized on a much more larger scale. Across all these methods, the most important variable was the parent SIF observation (usually coarse) when one exists followed by secondary predictor variables like optical variables like reflectance, NIRv, radiation variables like PAR, fPAR, land surface temperature, vapour pressure deficit and soil moisture content. Parent SIF was only available in continuous format for TROPOMI and GOSIF [LX19] and some of the kriging methods could use spatially sparse OCO-2 samples, but this was only reliable when their regional cluster density was high. Land cover classification and crop composition which was available at high spatial resolution was also found to be important for decomposing the coarser SIF signal to a finer scale. Majority of the methods used a tabular model as reliable large sample training for the convolutional models can be challenging. Both tabular and convolutional models were able to reliably enhance coarse SIF measurements, with tabular methods often including neighbouring information through feature engineered spatial factors. The CNN methods [GTK+22], [DXH+25], [ZXQL20] could naturally learn neighbourhood patterns, field boundaries and additional spatial context that cannot be reliably represented in tabular format. However a CNN model cannot recover physiological information that is absent from its predictors. Therefore predictor quality, spatial support and external validation design mattered more than choosing a convolutional or tabular model. No single study tested if model architecture matters when provided with the same data, spatial and temporal scope, therefore an an interesting point of research could be to address this.

Using OCO-2 as the base target support (unlike TROPOMI, GOME-2) can be challenging especially in gap filling contexts because of its narrow recording track, spatially wide swaths between each track. For a single year OCO-2 SIF data, there will be large regions without a parent OCO-2 SIF data to draw any context of general magnitude of

SIF in this region. Though this can be mitigated by combining multiple year data at the same location, but in turn lose year specific characteristics. One major research gap is that majority of the methods output enhanced SIF is at 0.05 resolution with a few methods reporting 500m, no SIF product exists at a lower resolution than 500m. For crop yield studies in regions with small farm sizes (usually under 60 hectares), a higher than 500m resolution SIF product is necessary to acquire crop pure SIF signals. Most of the models highest resolution predictors were the MODIS reflectance bands at 250m-500m resolution. Sentinel-2 provides reflectance band data in 10m, 20m, 60m resolution across its 13 bands. As per the literature review, OCO-2 reconstruction with very high resolution predictors like Sentinel-2 predictors has not been attempted and therefore remains a promising research direction. It also remains insufficiently tested whether convolutional models outperform tabular models when both use the same Sentinel-2 information. As previously discussed, SIF offers a higher crop yield predictor power than other secondary data, therefore it remains to be tested if isolating crop pure SIF from mixed SIF footprints can support more accurate crop yield estimations [QLH+22, SGG+18, PGZ+20].

Table 3.2: Summary of remote-sensing-based crop yield, crop productivity, and GPP studies. All RMSE values are reported in t ha^{-1} units

Reference	Prediction target	Remote-sensing inputs	Additional predictors	Model	Study area	Period	Performance
SIF-based yield, productivity, and stress estimation							
Peng et al. [PGZ+20]	Maize / Soybean	GOME-2/downscaled SIF; NDVI; MODIS GPP	County effects	Fixed-effects regression	United States; county	2015–2018	$R^2 = 0.85 - 0.89$; RMSE = 0.85
Guan et al. [GBZ+16]	Maize / Soybean	GOME-2 SIF	C3/C4 type; respiration; temperature	Process-guided conversion	United States; county	2007-2012	$R^2 = 0.43 - 0.72$
Zheng et al. [ZHN+25]	Winter-wheat yield	SIF; NIRv, NDVI, EVI, EVI2	Climate and soil	XGBoost; RF; ANN; SVM	North China Plain; municipality	2007-2020	$R^2 = 0.87$; RMSE = 0.352
Xue et al. [XHL+25]	Wheat production	Satellite SIF	Temperature; soil water; VPD	Process-guided q_L model	Australia; state	2019-2022	$R^2 = 0.86$-0.91
Kira et al. [KWH+24]	Maize / wheat	OCO-2 SIF; MODIS NDVI, NIR, fPAR	PAR; NPP:GPP; area; harvest index	MLR; ANN; RF	U.S. counties; Indian districts	2015-2020	$R^2 = 0.53$-0.69
He et al. [HMD+20]	Maize / Soybean	PhotoSpec and TROPOMI SIF	GPP; planted area; C3/C4 fractions	Cross-scale regression	U.S. Maize Belt; tower/county	-	$R^2 = 0.72$-0.89
Wang et al. [WYL+25]	Maize / Soybean yield	Satellite SIF	Temperature; VPD; soil moisture; phenology	Process-guided q_L model	U.S. Midwest; county	2018-2023	$R^2 = 0.76$-0.81
Qiu et al. [QLH+22]	Maize / Soybean	GOSIF; GOME-2 SIF; NDVI, EVI, NIRv	Tower GPP; yield statistics	Anomaly regression	U.S. Midwest; regional	2008-2018	$R^2 = 0.91$ for GOSIF
Joshi et al. [JPCB23]	Winter-wheat yield	SIF and three VIs	Climate and soil	XGBoost; RF	CONUS; county	14 years	$R^2 = 0.69$
Song et al. [SGG+18]	Wheat heat response	SIF, SIF yield, NDVI, EVI	Temperature; absorbed light; yield	Anomaly comparison	Indo-Gangetic Plains	2010	Yield loss: observed 6%; SIF 13.9%
Optical and spectral-index yield estimation							
Vannoppen et al. [VG21]	Winter-wheat yield	Maximum NDVI	Monthly precipitation	Random forest	Belgium; field	2016-2018	$R^2 = 0.66$
Hunt et al. [HBC+19]	Wheat yield	Sentinel-2	Weather; terrain; soil moisture	Random forest	United Kingdom; 10 m/field	-	RMSE = 0.63
Vallentin et al. [VHI+22]	Cereal	Six sensors; 15 spectral indices	Timing; soil; terrain; field properties	Correlation analysis	Northeast Germany; field	13 years	$r \approx 0$-0.94; median 0.19
Chandel et al. [CTS+19]	Wheat yield	NDVI, NDWI, SAVI, NDNI	Nitrogen treatment; growth stage	Stage-specific regression	Central India	2014-2017	$R^2 = 0.95$-0.96
Multi-source and temporal yield models							
Goihl et al. [Goi24]	Six crop yields	-	Weather; historical yield	Random forest	Saxony	2015-2022	Wheat RRMSE = 2.88-20.56%
Marszalek et al. [MKS22]	Winter-wheat yield	Sentinel-2 bands; NDRE, REIP, NDWI	Rainfall; evapotranspiration	Linear regression; RF	Southern Germany; field	2016-2018	$R^2 = 0.84$; RMSE = 0.56
Brandt et al. [BBB+24]	Three crop yields	Multi-source EO variables	Weather; soil; parcels	Six-model ensembles	Two German states; parcel/district	2019-2022	$R^2 = 0.66$-0.89; nRMSE = 7.2-16.9%
Zhao et al. [ZPZ+20]	Wheat yield	Sentinel-2 OSAVI, NDVI, CI, NDRE	Crop-model water stress	Combined regression	Northeast Australia; field	2016-2017	Test $R^2 = 0.93$; RMSE = 0.64

Continued on next page

Table 3.2 - continued

Reference	**Prediction target**	**Remote-sensing inputs**	**Additional predictors**	**Model / approach**	**Study area / scale**	**Period**	**Reported performance**
Srivastava et al. [SSK+22]	Winter-wheat yield	-	Weekly weather; soil; phenology	1D CNN	Germany; county	1999-2019	RMSE reduction = 7-14%
SIF-based GPP estimation and evaluation							
Hu et al. [HM20]	Regional GPP	GOME-2 SIF; MODIS NDVI, fPAR, LST	Soil-moisture index	Moving-window regression	Flux-tower regions	-	-
Ma et al. [MGW+25]	Global GPP	Three SIF-derived GPP sources	Eddy-covariance GPP	Transfer learning	Global; tower network	-	R^2: 0.036-0.132
Shekhar et al. [SBG22]	GPP evaluation	CSIF, GOSIF, LUE-SIF, HSIF; VIs	FLUXNET/ICOS GPP	Statistical comparison	Global/Europe; tower	2007-2018	SIF generally exceeded NDVI/EVI

4 Research question

4.1 Problem statement

OCO-2 satellite SIF data provides information related to photosynthetic activity and in turn crop yield estimation, but being spatially and temporally sparse affects its direct usage for spatially continuous and field scale agricultural analysis. Existing enhanced SIF products are limited to 500m or coarser resolution and thus these SIF footprints can still contain SIF signals mixed from several crops and other vegetation. It is therefore necessary to construct a much higher resolution SIF product using high resolution predictors primarily from Sentinel-2 and test how model architecture, input representation, spatial aggregation, sample density and agro climatic variation can affect resulting predictions. The agricultural value of this enhancement must then be assessed by testing whether crop pure SIF improves winter yield prediction over a mixed SIF model.

4.2 Research Questions

RQ1: How accurately can OCO-2 Solar-Induced Fluorescence (SIF) be predicted from high-resolution spectral and climatological variables using machine learning, and how does the resulting enhanced SIF compare with existing state-of-the-art enhanced SIF products?

RQ2: How does the choice of machine learning architecture (tree-based models such as Random Forest and XGBoost versus neural-network-based models such as Convolutional Neural Networks) and the representation of input data (tabular features versus spatially structured CNN-compatible inputs) affect the accuracy and spatial consistency of enhanced SIF derived from OCO-2 observations?

RQ3: To what extent does spatial aggregation affect the stability and predictive accuracy of enhanced SIF derived from spatially and temporally sparse OCO-2 observations?

RQ4: How does the performance of the OCO-2-based SIF enhancement model vary across different agro-climatic zones within Germany?

RQ5: To what extent does high-resolution, crop-pure enhanced SIF improve winter wheat yield-prediction RMSE compared with mixed-pixel SIF derived from raw satellite observations?

RQ6: How does the spatial density of OCO-2 training samples affect the uncertainty of the reconstructed SIF product at the field scale?

RQ7: To what extent can the yield model trained on winter wheat be transferred to other C3 crops without loss of physiological signal integrity?

5 Methods

5.1 Research Design and Study Area

5.1.1 Overall Research Design

The thesis study followed a predictive research design that links satellite based SIF enhancement with an agricultural yield application. Rather than designing the work as a single end to end model, it was modeled in two analytical stages. The first stage uses machine learning based models to learn relationships between OCO-2 SIF observations and spatially complete biophysical, radiation, seasonal and crop-type predictors. The second stage uses the enhanced SIF data produced by the best performing model; Spatial Aggregate (4km) restricted to winter wheat pixels which was then grouped and summarized by month and NUTS level 3 administrative regions and finally used as a predictor for the winter-wheat yield model. To reiterate, SIF data was first spatially enhanced, given completeness spatially and temporally and this enhanced SIF data is finally used to test if it could provide useful signal for a crop yield prediction model. Both of the stages used historical predictor data with held-out test or inference data which ensured the results describe a predictive association between the target and predictor variables rather than a casual effect of an individual predictor.

The SIF enhancement stage was designed as a supervised mapping between a set of spatial predictors to a SIF value with a defined spatial support (spatial resolution of the predictors). We can define this mathematically as

$$\widehat{S}_{u,t} = f_\theta(\mathbf{X}_{u,t}), \tag{5.1}$$

where f_θ represents a fitted tree-based, multilayer-perceptron, or CNN model with parameters θ, u denotes the spatial support or OCO-2 footprint, t the observation time, $\mathbf{X}_{u,t}$ the predictor variables, and $S_{u,t}$ the OCO-2-derived target (this input target representations can vary from a single data point to spatial aggregates combined multiple targets into one value, this will be discussed in depth in **??**). For area-to-point MLP, $\mathbf{X}_{u,t}$ was a footprint-level matrix containing the predictor values for every 20 m pixel, For polygon-mean models, $\mathbf{X}_{u,t}$ was a tabular summary of footprint-level statistics, For CNN models, $\mathbf{X}_{u,t}$ was a multi-channel raster chip.

The SIF enhancement experiments used different model architectures and data representations (both target and predictors). For the polygon-mean models, mean aggregation of predictor values falling within an OCO-2 footprint were averaged and fitted to a single SIF value, Area-to-point model uses a multilayer perceptron to fit pixel level sentinel-2 data to footprint scale SIF. Single footprint CNN experiments uses one target footprint in one image chip (channels representing predictors), whereas multi footprint CNN experiments uses multiple same date footprints in one image chip after which a footprint mask-averaged prediction was computed. The best experiment; spatial-aggregate (4km) combines by averaging at least 5 OCO-2 observations in a fixed 4km grid cell and a CNN model is trained on it with an aggregate footprint-weight map as many of the 4km grid cells will have some it's SIF polygons falling partially outside the cell. Given that almost all of the experiments have different architectures and data representations, if one model performs better, the improvement cannot be attributed only to it being a CNN, MLP, Random Forest, or XGBoost model. The result reflects the entire combination of data preparation, predictors, target definition, spatial support, and model architecture.

The downstream task of predicting winter wheat yield used the same predictive logic as the first task, but works on a much more coarser spatial scale. Since the yield data available is at NUTS 3 level which is the highest resolution, we can only conduct the yield analysis at this spatial scale. For NUTS-3 region r and year y, the winter-wheat yield can be represented as

$$\widehat{Y}_{r,y} = g_{\phi}(S_{r,y,3}, S_{r,y,4}, S_{r,y,5}, S_{r,y,6}, S_{r,y,7}), \tag{5.2}$$

where $S_{r,y,m}$ denotes a monthly SIF predictor for March through July and g_{ϕ} denotes the fitted yield model. To see if the SIF enhancement helps improve the yield modelling, separate versions of the input data were produced in the form of crop pure enhanced SIF (ie; only winter wheat pixels' SIF values were taken), raw OCO-2 footprint SIF values which is a representation of mixed crop SIF values. The raw OCO-2 SIF values was also constrained to SIF polygons containing at least 10% winter wheat out of the SIF footprint crop area. The yield model design held the yield target, region-year groupings and Random Forest configuration constant while changing the SIF representation. To reiterate, the crop pure predictors where SIF values taken from crop pixels that were winter wheat, whereas the raw baselines took OCO-2 footprints with either no crop purity requirement or a minimum crop (winter wheat) coverage threshold (5%, 10%, 30%). Years 2019-2022 subset was used for model training and 2024 subset was used as the test data. This ensured any performance evaluation can be said to be generalized over years, ie; it measured whether crop-targeted enhancement reduced out-of-year prediction error compared to SIF observations that were combination of different crop SIF values for a given footprint.

5.1.2 Geographical Scope

The SIF enhancement study domain was Germany, where the data domain was represented by a set of Sentinel-2 Military Grid Reference System (MGRS) tiles rather than by a single national level raster. Since the Sentinel-2 data (the main high resolution predictor data) was only available in MGRS tiles format, we have to stick to tile based data domain rather than state boundary, agricultural zone etc. based domain. The nine MGRS tiles used for the Sentinel-2 training predictor data were `T32ULA`, `T32ULB`, `T32UMD`, `T32UNA`, `T32UNC`, `T32UNE`, `T32UPC`, `T32UPE`, and `T32UPU`. These tiles sampled data across north, centre and south of Germany and included all the major crop-producing states in Germany: Bayern, Niedersachsen, Sachsen-Anhalt, Mecklenburg-Vorpommern, Nordrhein-Westfalen, Rhoilond-Palz, Hessen. Figure **??** shows the distribution of SIF polygons from 2019-2024 in the 9 MGRS tiles. Since the experiment design has different model architectures and data representations, we cannot claim an uninterrupted high resolution coverage of every location in Germany for every experiment as the predictions are restricted by availability of OCO-2 tracks, Sentinel-2 product availability, quality filters and selected months in the input data. Therefore any Germany-wide conclusions can only refer to the diversity represented by the selected tiles and observations rather than to an exhaustive census of all German pixels.

Two additional MGRS tiles, `T32UQV` and `T32UMC`, were used for external native-footprint evaluation of the best model; Spatial-aggregate (4km) model. This helped quantify if target to predictor relationships learned on the 9 MGRS tiles could be transfered to new locations or in this case, new MGRS tiles. Also the external native-footprint evaluation tests whether a 20m resolution SIF map predicted by the model and then aggregated to the native OCO-2 footprint matched with the original OCO-2 footprint SIF value, if it tested well then we can conclude that both geographic transfer to unseen tiles and transfer from aggregated training support back to approximately native OCO-2 footprint support works well.

The winter-wheat yield model focused on Bavaria as it is the largest wheat producing state in Germany and making 20m resolution SIF maps was extremely computationally and time consuming. Fives MGRS tiles were used for the winter-wheat yield model in Bavaria: `T32UNA`, `T32UPU`, `T32UQV`, `T32UPA`, and `T32UPV`. A Bavarian NUTS 3 region was retained if at least 50% of it's area fell within one of the above MGRS tiles. Visual inspection of the NUTS 3 regions as seen in Figure **??** and associated winter wheat mask map as seen in Figure **??** helped confirm that the NUTS 3 regions that were only partially present in the MGRS tile had majority of their wheat producing areas in the MGRS tile. Yield prediction was done at NUTS 3 region-year level because this was the highest spatial and temporal availability of the official yield data.

Agro-climatic variation in the form of Plant Hardiness zones was represented independently of the administrative and raster tiling data domains. Each SIF observation inherited one or more plant hardiness zone labels from the spatial zone layer, and model performance was later summarized for the individual zones 7a, 7b, 8a, 8b, and 9a. Mixed labels occurred due to some of the SIF polygons falling across multiple plant hardiness zones. MGRS tiles, federal states, NUTS regions, and hardiness zones therefore served different purposes and were not treated as interchangeable regional definitions, each are suited to how the data was available. MGRS tiles defined Sentinel-2 raster storage and experimental partitions, federal states gave geographic context, NUTS 3 regions was required for the yield target support and plant hardiness zones described agro-climatic conditions. Keep these geographic roles separate allowed a fitted SIF enhancement model to be evaluated by state boundary geography, season, agro-climatic zone without changing its corresponding prediction target.

5.1.3 Temporal Scope and Spatial Supports

The main SIF enhancement predictor and target datasets covering year range 2019-2024 focused on months February to July as these are the main growing period for Winter Wheat in Germany. OCO-2 observations are basically tied to individual acquisition dates, the OCO-2 satellite do not make regular revisit over the same track instead are staggered by weeks to months around the track and therefore highly irregular in time. Sentinel-2 predictors are monthly composites whose composite temporal range can vary from 7 days to 45 days. GLASS products other than GLASS PAR are 8-day composites, eg; Day of Year 01 composite (DOY 01) covers DOY 01 to DOY 08. GLASS PAR is available as daily source files. VIIRS PAR is represented as daily mean source files. Dates were matched according to the available date range of each product rather than with one common nearest date rule. The predictor variables were temporally matched with the OCO-2 observation dates in the following method: Sentinel-2 records were linked by month and product interval, eight-day GLASS products were linked to the interval containing the SIF date, and VIIRS PAR was matched by exact date as it is a daily source file. OCO-2 records that were outside a Sentinel-2 product interval as obtained from the supporting WASP DTS raster files (these files were available for 10m and 20m reflectance bands and contained days since start of year for each pixel, thus essentially allowing us to identify the temporal range of the monthly composite) were retained with a `date_align` flag for later evaluation. The yield analysis used March to July in 2019-2022 and 2024, while 2023 was excluded because Sentinel-2 WASP products had poor coverage for this year. Years 2019-2022 were used for model development and 2024 was held out for testing. The predictors were kept as separate month wise average predictors allowing to preserve any seasonal information in winter wheat growth and senescence.

The thesis study's experiments developed several spatial supports (target and predictor spatial resolution) that were kept distinct for each modelling and evaluation experiment. Sentinel-2 bands at 10 m and 20 m were aligned to a common 20 m grid to better align spectral index calculation and also reduce computational complexity. Every other data source would later be aligned to this common 20m grid as in the case of the CNN models since they require consistent image resolutions and dimensions for training. The 10m crop type rasters were converted to fractional crop maps on the common 20m grid. GLASS FAPAR had an actual pixel spacing (resolution) of approximately 231.7 m while VIIRS PAR had a spacing of approximately 927 m, these were both aligned with the Sentinel-2 predictors. Each 20m pixel would acquire it's corresponding GLASS predictor value depending on the alignment, many 20m pixels can have the same GLASS values due to the considerable resolution differences. OCO-2 observations were irregular polygons ranging from elognated parallelograms shape (Nadir) to triangle, trapezoid shape (Glint) with areas typically close to 1 km^2. Single-footprint and multi-footprint enhancement experiments preserved the individual footprint targets, but as the name suggests, multi-footprint chips included several separate masks (each mask corresponding to a SIF footprint). Spatial-aggregate (4km) experiment averaged at least five same-date observations within a fixed 4 km cell represented by 200x200 predictor pixels (each pixel at 20m resolution). External Native OCO-2 resolution inference tested the Spatial-aggregate (4km) model by averaging its 20 m predictions over individual OCO-2 footprints in unseen tiles and comparing the averaged value to the original SIF observation value. For the winter wheat yield model, Spatial-aggregate (4km) model predictions over each 10m winter wheat pixels (regridded to 20m) were aggregated to one monthly value for each NUTS-3 region and year. The Spatial-aggregate (4km) model produces SIF predictions at 20 m, but since there is no real 20m SIF data for validation, the validation could only be done at Native OCO-2 SIF resolution (around 1 km^2). The 20m SIF predicted map is thus a downscaled estimate and not a directly verified 20m ground truth, more will be discussed in the Results section.

5.2 Datasets

5.2.1 OCO-2 SIF Observations

NASA OCO-2 observations provided the target variable for all the SIF enhancement experiments. The data source was the Level 2 bias-corrected solar-induced fluorescence product generated with the IMAP-DOAS retrieval and distributed as daily files planet wide coverage by the NASA Godard Earth Sciences Data and Information Services Center (GES DISC). Two product releases V11.r and V11.2r were combined row-wise and used for the thesis study [OPC20, OPC24]. The files contained daily-corrected SIF retrievals at 740 nm, 757 nm, and 771 nm, together with their retrieval-uncertainty fields, acquisition time, latitude and longitude, four latitude & longitude corners describing the elongated ground footprint (usually in the form of an elongated parallelogram), quality flags, measurement mode, viewing geometry, illumination geometry, source-file identifiers, and selected meteorological fields. Measurement mode determined if the observation was retrieved in Nadir or Glint mode. Quality Flag determined the quality of the SIF observations with higher number flag given to observations obscured by cloud, haze and other climate conditions, consequently Quality Flag (0,1) was chosen for this study. Phase angle was calculated from the viewing geometry and illumination geometry and helped determine if SIF values drastically changed in a given area if the phase angle changed considerably. The temporal range for OCO-2 observations for this study covered 2019-2024. Even though years preceding and following this range existed, due to the temporal availability of year wise crop type rasters (2019-2024), the study's data temporal range was restricted to 2019-2024.

5.2.2 Sentinel-2 L3A WASP Monthly Composites

The driving predictor data for high resolution SIF enhancement was Sentinel-2 L3A WASP data, these were monthly composites Sentinel-2 composites produced by the Weighted Average Synthesis Processor (WASP). WASP forms temporal syntheses from atmospherically corrected Sentinel-2 Level 2A inputs generated with MAJA, with the processing specifications documented by Hagolle et al. [HHD+17, HMK18]. The WASP products (product referring to one year-month composite data directory containing the various Band raster and supporting files) were organized by MGRS tile, year and monthly synthesis date which provided consistent spatial reference for the study areas in Germany. The monthly composite product supplied blue, green, red, and near-infrared reflectance bands B2, B3, B4, and B8 at 10 m resolution. Red-edge, narrow near-infrared, and short-wave-infrared bands B5, B6, B7, B8A, B11, and B12 were available at 20 m resolution. Each product also contained 10 m and 20 m classification layers, identified as `FLG_R1` and `FLG_R2`, alongside the Band rasters which helped determine if the data pixel was land, water, cloud, snow or no data. Band (reflectance) value was encoded as signed 16- bit integers with a quantification value of 10,000, with no data assigned as -10,000. The temporal coverage for the data was monthly composites for February to July during 2019-2024. Since 20m resolution was still a high resolution, 10m reflectance bands were aggregated to 20m resolution in the data preprocessing steps since all the input data needed to be at the same spatial resolution and a lot of computation power was gained with not dealing with 10m resolution. Each year-month composite collection was around 30-35GB in size and took around 30-50mins to download from Centre national d'études spatiales (CNES) GEODES portal. An earlier WASP collection from Deutsches Zentrum für Luft und Raumfahrt (DLR) EOC service was available for download, but data preceding 2024 were archived in May, 2026 and was no longer available for download.

5.2.3 GLASS Vegetation and Radiation Products

The Global LAnd Surface Satellite (GLASS) product suite provided medium resolution vegetation index and radiation variables. Regular updates and additions of new spatial data to the GLASS product suite have been done from 2013-2023 as shown in [LZL+13, LZX+13, LCJ+21, LCC+23]. Before an expensive model data preparation and model training could be done on very high resolution Sentinel-2 predictors (20m resolution), the GLASS provided predictors was used to test if the overall data preparation methodology and model architectures were suited for the SIF enhancement task. The GLASS products were fraction of absorbed photosynthetically active radiation (FAPAR), photosynthetically active radiation (PAR), normalized difference vegetation index (NDVI), and enhanced vegetation index (EVI). FAPAR was obtained from the GLASS09D01 V60 archive at 500m resolution, PAR was obtained from GLASS04B01 V42 at 0.05° spatial resolution while NDVI and EVI were obtained from the corresponding GLASS13D01 and GLASS14D01 250 m MODIS products. These products other than PAR were supplied as eight-day composites on the MODIS sinusoidal grid and stored in HDF format. MODIS sinusoidal tiles `h18v03` and `h18v04` were the two tiles that covered the entirety of Germany. The temporal range was 2019-2024 and focused on composite start days from day of year 033 to 209 (covering February to July approximately). PAR in contrast was supplied as daily source files, but for the case of this study, eight day sequence matching FAPAR, NDVI and EVI day of year were downloaded to reduce the amount of data to be downloaded and processed. These GLASS products will also be referred to as Moderate Resolution

Imaging Spectroradiometer (MODIS) variables as they are enhanced data derived from raw spectroradiometer data as given in FILLREF.

5.2.4 VIIRS Daily PAR Product

Daily Photosynthetically Active Radiation (PAR) data for the principal Sentinel-2 workflow came from the VIIRS/NPP Photosynthetically Active Radiation Daily Level 3 Global 1 km Sinusoidal Grid Version 2 product, identified as VNP18A2 [Wan25]. This data has a much higher resolution (1km) than the GLASS provided PAR data at 0.05°. The product provides daily average photosynthetically active radiation and it's associated categorical quality layer. Its native grid is a VIIRS sinusoidal projection with a pixel resolution of approximately 927m. This sinusoidal project or coordinate reference system was later realigned with the optical predictor grid as described in Section 5.3. The quality layer provides quality codes that describe the data's usability rather than a continuous numerical ranking between the retrievals. Daily mean PAR was chosen instead of Instantaneous PAR (available at every 3 hours intervals daily) because the OCO-2 SIF values were daily-corrected SIF values. PAR data was presented in the form of HDF files which were downloaded for the unique OCO-2 observation dates. Like the FAPAR dataset, Germany scope data was present in two separate tiles which had to be mosaicked properly.

5.2.5 Crop, Regional, and Yield Data

Annual crop type raster was obtained from the German Aerospace Center (DLR) EOC Geoservice [AGAG+22, GHA+25]. The yearly rasters provided a 10m resolution categorical raster that marked and labelled every classified crop pixel on Germany's map into 21 distinctive crops falling in the crop groups: cereal, oilseed, root-crop, grassland, woody-crop, other agricultural, and non-crop classes. These annual crop type rasters provided the crop spatial information needed for footprint level crop composition predictor variables and downstream winter wheat yield analysis. Administrative boundaries of Germany, German federal state boundaries, NUTS 3 regions were obtained from the Geographic Information System of the Commission through the `giscoR` interface [Her26]. NUTS level 1 represented German federal states, while NUTS level 3 represented the districts and independent cities used in the Bavarian yield analysis. German plant hardiness zone polygons defining the agro-climatic regions for Germany were acquired by downloading the geojsons in the leaflet data directory provided by [Pla26]. Official Bavarian crop harvest tables provided district-level yields for 2019, 2020, 2021, 2022, and 2024 [Bay26]. Year 2023 was not included in the yield model study because Sentinel-2 WASP products had limited temporal coverage for 2023, ie; not all the necessary months were available for all previously defined MGRS tiles. The target variable for the yield model was *Winterweizen (einschl. Dinkel und Einkorn)* which reports winter wheat yield. The yield values were reported in decitonnes per hectare using German numeric formatting, these would be converted to tonnes per hectare to allow for cross-study comparison.

5.3 Data Preprocessing

5.3.1 OCO-2 Target and Footprint Preparation

The downloaded OCO-2 files were converted into a row-wise observation table where each row represented one SIF observation (sounding). The main target variable was constructed from the daily corrected SIF values at 757nm and 771nm. For sounding i, the combined target as noted in FILLREF was defined as

$$S_i = \frac{S_{757,i} + 1.5 S_{771,i}}{2}, \tag{5.3}$$

where $S_{757,i}$ and $S_{771,i}$ are the daily-corrected retrievals in the two oxygen absorption bands. 771nm SIF retrieval is multiplied by 1.5 to bring it closer to the scale of 757nm retrieval before we can combine both the values into one SIF value is that much more noise free than the individual retrievals. This combined target is called `target_modis_sif` in the processed SIF data. The 740nm retrieval target was used for the initial experiments, MODIS 16×16, area-to-point as seen in FILLREF. As mentioned earlier, every OCO-2 SIF retrieval comes with an uncertainity factor, retrieval uncertainity was adjusted using the OCO-2 daily correct factor before any unreliable observations could be removed. Let $\sigma_{757,i}$ and $\sigma_{771,i}$ denote the corrected uncertainties, the uncertainty of the combined target was calculated as

$$\sigma_i = \frac{1}{2}\sqrt{\sigma_{757,i}^2 + (1.5\sigma_{771,i})^2}. \tag{5.4}$$

In this study, negative SIF was not automatically removed eventhough theoretical SIF value cannot be negative. This is because small negative retrievals can occur during observation when the signal is close to zero. A negative value was accepted when $S_i + 2\sigma_i \geq 0$ and marked as questionable when $S_i + 2\sigma_i \geq 0$ check failed and $S_i + 3\sigma_i \geq 0$ was satisfied, and finally the negative value was rejected when $S_i + 3\sigma_i < 0$ was satisfied. The main datasets for the experiments kept negative SIF values that were classified as accepted by this rule. After this, a broad SIF bin range check of $-0.5 \leq S_i < 2.0$ was then used to remove implausible combined values. Quality Flag and Measurement mode was restricted to values 0,1. For Quality Flag this represented best and good values, whereas for Measurement mode this represented Nadir mode and Glint mode.

Each SIF sounding (observation, row) was built into a polygon from its four recorded corner coordinates instead of using the existing latitude and longitude, as the latter were found to be offset by some distance from the polygon's true centroid. This would complicate later SIF polygon assignments, alignments to various grids, boundary selections. The corner pairs were ordered from the original matrix representation to form a closed polygon in WGS 84 coordinate reference system (CRS), and any invalid polygons formed during polygon construction were repaired if possible. For this study, polygon area and centroid operations were carried out after transformation to the coordinate system ETRS89-LAEA Europe (EPSG:3035), so that distances and areas were measured in metres as the WGS 84 CRS is for latitude and longitude representation. The centroid retrieved from the polygon was used for assignment to federal states, plant hardiness zones, MGRS tiles, fixed grid cells, and NUTS-3 regions. The polygon geometry was important for raster extraction and for fractional footprint masks. This was important as an elongated OCO-2 footprint can cross a raster or administrative boundary even when its centroid lies on only one side. Every SIF observation row also kept a row identifier, acquisition date, source file. These metadata fields allowed SIF observations to be traced through later joins, chip manifests, model predictions, and any subgroup evaluations.

5.3.2 Temporal Matching and Spatial Alignment

Most of the data products as seen in FILLREF do not have exact temporal alignment or matching, therefore the predictor dates were matched according to the temporal meaning of each data product. An OCO-2 SIF observation was matched to the Sentinel-2 monthly composite for the same tile, year and month. As mentioned earlier, each Sentinel-2 composite comes with its own DTS raster which contains the lower and upper bound date acquisition ranges of each pixel in a reflectance (band) raster, these lower and upper bound dates were compared with the OCO-2 observation date. SIF observations inside this interval were labelled as `inrange` while observations outside were labelled `outrange`. Both date alignment groups were retained in the main dataset because most of the out-of-range differences were small and their performance could be reported separately FILLREF. The GLASS products eight-day FAPAR, EVI, and NDVI datasets were matched by the composite interval containing the SIF day rather than by the numerically closest composite start date. For example, a SIF observation on day of year DOY07 belongs to the product beginning on DOY01, since DOY01 composite describes days 01-08. Daily VIIRS PAR was matched to the exact OCO-2 date as they were available as daily source files, while the earlier GLASS PAR product used the closest available day to a SIF date.

In regards to spatial alignment, for the CNN models all image channels within one convolutional sample (chip) had to share the same rows, columns, extent, and coordinate reference system. The experiments using Sentinel-2 data thereby used a 20m reference grid in the UTM coordinate system. Sentinel-2 10m reflectance bands were reduced to this 20m grid by area averaging, while reflectance bands already at 20 m did not need readjustments. Each convolutional sample (chip) cropped its own GLASS FAPAR and PAR section and reprojected it to the 20m grid using bilinear interpolation. For VIIRS PAR, the PAR quality mask was applied on the native VIIRS grid before reprojection, and only quality codes 1 and 2 were accepted. For GLass products, FAPAR values outside the range 0 to 1 and PAR values outside the range 0 to 700 were treated as missing data. APAR was calculated as the pixel-wise product of FAPAR and PAR.

Categorical information had to be handled in a different way compared to the previous continuous reflectance and radiation (FAPAR, PAR, APAR) data. The Sentinel-2 classification masks provided by the FLG rasters were used to keep pixels that were only land recorded values, this was done before any of the spectral indices were calculated. Annual crop codes were converted into separate binary membership rasters and then averaged onto the 20m grid, this would be used the crop channel for the CNN model input chips. For the crop channel a value of 0.75 means that three quarters of the contributing 10m area (croptype rasters are originally at 10m resolution) belonged to the stated crop group. Since OCO-2 SIF polygons were vector data, they were rasterized as fractional masks, so boundary pixels could take values between zero and one according to the proportion of the pixel covered by the SIF polygon. Fractional masks were preferred to binary masks because a 20m (Sentinel-2 experiments) or 250m pixel (modis predictors, for the experiments FILLREF) rarely follows the slanted edge of a SIF polygon exactly, ie; the SIF edge polygon almost always diagonally cuts a pixel.

5.3.3 Predictor and Crop-Channel Construction

This section will present how the various channels for a CNN model input chip (image) were constructed. For this study, five spectral indices were calculated from land masked Sentinel-2 reflectance bands. They are defined as

$$\begin{aligned} \text{NDVI} &= \frac{B8 - B4}{B8 + B4}, & \text{NDMI} &= \frac{B8 - B11}{B8 + B11}, \\ \text{NDRE} &= \frac{B8A - B5}{B8A + B5}, & \text{NIRv} &= B8 \times \text{NDVI}, \\ \text{EVI} &= 2.5\frac{B8 - B4}{B8 + 6B4 - 7.5B2 + 1}. \end{aligned} \tag{5.5}$$

where B2, B4, B5, B8, B8A, B11 are reflectance bands. The reflectance values were divided by 10,000 before the spectral indices calculations were done and any reflectance value of -10,000 was treated as no-data. A small denominator tolerance prevented division by values close to zero for EVI, therefore EVI values outside the interval -1 to 1 were removed because the value can become extremely large when its denominator approaches zero. For the CNN models, these index calculations produced spatial channels for the input chip image rather than footprint averages. The same index definitions were also used for the polygon means experiments. Missing raster pixels were kept as `NaN` rather than being replaced by a meaningful number. This would help simplify the data normalization step where the predictor means and standard deviations were calculated from the finite training pixels only. After normalization, any remaining missing (`NaN`) pixels were replaced by zero, which represents the training-channel mean in the normalized space. This prevented raw zero from being confused with valid zero FAPAR, spectral index, or crop fraction values.

The crop data had 21 unique crop labels and therefore had to be simplified into few crop channels that represented the main crop categories in the study area. The channels for the input chip were winter wheat, winter barley, winter rye, maize, grass and forage, woody crops, other crops, active crops, and non-crop land. The woody channel combined vineyards with fruit trees and other woody agricultural vegetation, the grass and forage channel combined the grassland and fodder crop classes. Active-crop fraction was made to signify which crop areas had active crop growth during a particular period and therefore its construction depended on both crop class and observation month. For example, winter wheat's main growth period was considered from February to July and again in October and November (next cycle of crops), while maize was active from May to October. The active crop channel was formed by summing the fractions of crop classes whose predefined growth period included the current month. With this active crop channel, we could present crop data in such a way that it represents the seasonal effect of crop growth. The annual crop map remained year specific so that rotations and changes between years were not replaced by one static landcover map.

The month predictor variable (obtained from the SIF observation date) was represented by two spatially constant channels rather than by integers 2 to 7 (February to July, the month interval for this study). For month m, the channels were calculated as

$$m_{\sin} = \sin\left(\frac{2\pi(m-1)}{12}\right), \qquad m_{\cos} = \cos\left(\frac{2\pi(m-1)}{12}\right). \tag{5.6}$$

This representation places December and January next to each other on a cycle/circle and avoids an artificial large distance between month 12 and month 1, as the characteristics of December is similar January. The main CNN model input chip (using Sentinel-2 as the base dataset) therefore contained 19 channels: five spectral indices, FAPAR, PAR, APAR, seven grouped crop fractions, active-crop fraction, non-crop fraction, and the two month channels. Spatial information in the form of Latitude and longitude were not included as image channels as the SIF data was extremely spatially sparse in a given year and therefore during the external inference testing stage there could have been test samples whose Latitude and Longitude were outside the training range for these two variables. Also excluding them encouraged the model to use environmental and seasonal information rather than directly memorizing OCO-2 SIF track locations (each SIF observation for a date is part of long date track as can be seen in FILLREF). During model training, the channel order was written to the dataset configuration and later checked against the model checkpoint during inference. This was important because changing channel order would silently associate trained weights with the wrong predictor.

5.3.4 Model-Ready Data Representations

The processed predictor variables were represented differently for tabular, area-to-point and CNN models. For the polygon mean model (tabular data), every SIF observation becomes one tabular row and the mean of predictor channel c was calculated from the pixels intersecting its footprint:

$$\bar{x}_{i,c} = \frac{\sum_{p \in i} a_{i,p} x_{p,c}}{\sum_{p \in i} a_{i,p}}, \tag{5.7}$$

where $a_{i,p}$ is the fractional area of pixel p inside footprint i. Each row in the data table contained one combined SIF target and one value for each spectral index, radiation and crop predictor, each value representing the mean value of the predictor values falling in one SIF footprint. APAR was calculated as mean FAPAR multiplied by mean PAR. The Random Forest and XGBoost models used this table which removed within footprint spatial structure (as the predictors were reduced to one mean value). The single footprint CNN model datasets centred one multi-channel raster chip on each SIF footprint and stored the SIF footprint as a fractional mask. The predicted SIF map (at predictor resolution) was averaged over this mask before comparison with the observed footprint value. The area-to-point MLP model data represented each footprint as a set of no more than 200 spatially distributed Sentinel predictor rows and assigned the same footprint level 740 nm target to those rows. Each OCO-2 footprint at 1km^2 contains as many as 2,500 Sentinel-2 pixels (20m). For the sake computational reduction, at most 200 pixels were sampled across the footprint, and their predictor values were given to the MLP. Because no pixel-level SIF observations existed, every sampled pixel shared the footprint's observed 740 nm SIF value. Its pixel level predictions were then averaged back to the footprint level for comparison to original observation value because real observed 20m SIF labels are unavailable (the best resolution for validation is OCO-2 SIF at 1 km^2). Therefore, the area-to-point MLP and MODIS 16 by 16 dataset differed from the principal models in both target wavelength (740nm vs combined SIF target) and filtering.

The Multi-footprint (MultiSIF) model datasets grouped several nearby same-date observations in one predictor chip by the means of density clustering keeping a minium of 4 SIF polygons and max range 10-11 polygons falling in one window, while retaining a separate mask and target for every footprint. The Sentinel-2 version used a 6km by 6km chip at 20m resolution, containing 4-10 footprints and each footprint having at least 90% of its area falling inside the chip. Its errors were averaged within each chip so that chips containing more footprints did not automatically have greater influence. The Spatial aggregate (4km) model dataset used a different target by assigning same-tile, same-date, and same-mode footprints to fixed 4 km cells containing at least five observations. Each cell had 200 by 200 predictor pixels, and its observed target was

$$\bar{S}_g = \frac{1}{n_g} \sum_{i=1}^{n_g} S_i, \tag{5.8}$$

where n_g is the number of assigned footprints and S_i is their observed SIF. For simplicity sake and to keep enough input chips for training purpose, the aggregate weight map gave equal influence to the normalized footprint masks even when their areas inside the chip differed. Since multiple footprints were being averaged down into one value, partial footprint coverage over the chip would not significantly change the accuracy of the overall mean. After this best model (Spatial aggregate (4km)) was selected for further validation, native-footprint inference chips were prepared for two unseen MGRS tiles using the same 4 km context, 20 m grid, and channel order. The resulting 20 m predictions were averaged over each observed footprint mask and compared to the original SIF observation value. It should be noted that these external samples were not used to refit the model or its calibration, they were only used to see if the model's strong training performance could be carried over to external data.

For the yield model dataset, Bavarian NUTS 3 regions were chosen in the selected five MGRS tiles if at least 50% of their area fell inside an available MGRS tile (Sentinel-2 data is represented in MGRS format), any duplicate tile assignments were removed, one tile was assigned to one NUTS 3 region depending on area overlap. The annual crop type rasters provided the winter wheat pixels spatial location for every retained NUTS 3 region, year and month from March to July. Since the CNN predictor grid was at 20m resolution, wheat map had to be converted to 20m resolution by required all four contributing 10 m crop pixels to be classified as winter wheat. The best model Spatial Aggregate (4km) was used to predict the high resolution SIF map only on the blocks containing these 20m wheat pixels and finally the wheat pixel SIF values were averaged to one crop pure monthly value for each NUTS 3 region. Raw comparison variables were formed from the OCO-2 SIF footprints and from subsets with at least 5%, 10%, or 30% winter-wheat coverage (eg; a SIF footprint has at least 30% of it's crop area as winter wheat). Footprints were first averaged within the acquisition date and then across the available dates in each month so that dense OCO-2 tracks did not dominate the regional predictor (some date tracks, have more SIF observations than other date tracks). The crop-pure and raw monthly predictor values were reshaped into one row per region and year and joined to official winter wheat yield table. The winter wheat yield table was formed by combining yearly NUTS 3 categorised yield files into one table. The yield unit was converted from decitonnes per hectare to tonnes per hectare so that it could be compared with external studies, and rows without a winter wheat yield (target value) were removed. Missing predictor values were replaced using means calculated from 2019-2022 only, so the held-out 2024 observations did not affect preprocessing. The processed table containing the SIF month wise averages and yield value allowed crop-pure enhanced SIF and mixed raw OCO-2 SIF to be compared using the same regional yield records.

5.4 Model Architectures

This section will only discuss the model architectures for the deep learning model experiments. The area-to-point model used a fully connected multilayer perceptron (MLP) that received one 20m predictor row at a time. Each row contained 17 features: Sentinel-2 spectral indices, crop indicators and two cyclical month variables. The input layer was connected to a hidden layer with 64 neurons, followed by a rectified linear unit (ReLU) activation and dropout with a probability of 0.1. A second hidden layer reduced the representation from 64 to 32 neurons and again applied the activation ReLU and dropout. The final linear layer of the MLP model converted the 32 neuron values into one pixel-level SIF estimate. No output activation was used because SIF was treated as a continuous regression target that could include values below zero. The model's MLP weights were applied to every sampled pixel independently, it did not know where the pixel was located relative to other pixels in the footprint or what its neighbouring pixels contained (ie; it has no spatial understanding of how the predicted SIF values are connected in a SIF footprint). Because OCO-2 provides one observed SIF value for the whole footprint rather than separate 20m observations, the model's pixel-level predictions inside that footprint were averaged to one predicted value. This predicted value was compared with the observed footprint level SIF during training to calculate the necessary performance statistics. Therefore, the model learned pixel-level relationships indirectly from footprint level supervision.

The single footprint and multi footprint CNN experiments used a compact U-Net architecture that preserved the spatial arrangement of the predictor channels. The models using MODIS variables (GLASS products) used two encoder levels with 32 and 64 feature maps, followed by a 128 feature map layer (the deepest representation layer of the model) and this was followed by two decoder levels with 64 and 32 feature maps, matching symmetrically. Each convolutional block contained two 3×3 convolutions, and every convolution was followed by batch normalization and a ReLU activation. Likewise the Sentinel-2 multi footprint CNN model (MultiSIF) used the same two-level layout as in single footprint model (MODIS) but started with 16 feature maps, 32 feature maps, and finally 64 feature maps in the deepest layer. Its convolutional blocks used group normalization and the SiLU activation instead of batch normalization and ReLU. This is because the MultiSIF model using Sentinel-2 predictors used small training batches and the 20m resolution image chips were much larger than the MODIS chips and therefore consumed more GPU memory. Batch normalization can become unstable with small batches because it estimates statistics from the current batch whereas group normalization calculates statistics within groups of the feature channels and is therefore less dependent on batch size. SiLU was used because it provided a smooth nonlinear activation and retained small negative responses instead of discarding them completely. In both experiments, 2×2 max pooling reduced the spatial dimensions in the encoder section and 2×2 transposed convolutions restored them in the decoder section. Skip connections concatenated the encoder features with the corresponding decoder features which allowed local boundaries and fine spatial patterns to be recovered back after pooling (pooling would condense these fine spatial patterns). Finally, a final 1×1 convolution produced one SIF value for every output pixel without applying an output activation. The predicted map was then reduced over either one footprint mask (single footprint model) or several separate footprint masks (multi footprint model).

The best model; Spatial aggregate (4km) used a deeper three-level U-Net compared to a two-level U-Net because its input contained 200 pixels by 200 pixels and could support an additional stage of spatial abstraction without losing too much information. A 200 by 200 output would go from 200, 100, 50, 25 after three pooling operations. The deepest layer still has a meaningful 25 by 25 spatial representation, whereas for the smaller 16 pixels by 16 pixels (single footprint) and 24 pixels by 24 pixels (multi footprint), three pooling stages would grealty compress any meaningful information away. The experiment's 19 predictor channels entered the encoder sections with 16, 32, 64 feature maps and final representation layer with 128 feature maps, the decoder section will reverse this sequence and finally a 1×1 output layer restored a single 200 by 200 predicted SIF map. Every block again used two padded 3×3 convolutions with group normalization and SiLU activations. Three max-pooling operations reduced the 200 pixels by 200 pixels input successively to 100 by 100, 50 by 50, and 25 by 25 pixels. The model produced a separate predicted SIF value for every 20m pixel in the output map and during training, the 20 m predictions were combined using the footprint-weight map and compared with the single aggregate target (4km). It should be noted that the 20m outputs are model derived predictions constrained by aggregate observations, rather than independently validated 20m measurements (since real 20m SIF observations do not exist).

5.5 Model Training and Experimental Design

Partitioning the model data into train, validation and tests was done before any data normalization, to avoid train statistics from leaking into the validation and test splits. Each model had different methods of partitioning the data (can call this as the partitioning or split unit). Different methods were chosen to see how it affects model performance with the final partitioning unit being used for the Spatial aggregate (4km) model. For the best model; Spatial aggregate (4km), input chips that shared either an OCO-2 acquisition date or a Sentinel-2 product were connected into one component, and these complete components were assigned uniquely to the training, validation or testing splits. This helped prevent the same OCO-2 acquisition date observations or predictor image (chips) from appearing in more than one partition. The components were allocated to an approximate 80/10/10 split by number of chips counts (not exact as it should satisfy the following conditions) while balancing month, measurement mode and MGRS tile composition across the three partitions. The Sentinel-2 multi footprint (MultiSIF) model also used a 80/10/10 split but its data partitioning unit was the complete acquisition date and balancing was done on month and measurement mode, ie; the data partitions contain a balanced distribution of the months and measurement modes. Unlike the Spatial aggregate (4km) model, this date based split did not prevent the same Sentinel-2 product from occurring in different partitions, so generalisation over MGRS tiles or regions cannot be ascertained fully with this experiment. The MODIS multi footprint model used year wise partitioning, 2019-2022 subset was used for training, 2023 for validation and 2024 for testing. Whereas the MODIS single footprint models used a 70/15/15 random stratification split by state, year and MODIS composite day. The area-to-point MLP model used 70/15/15 split and ensured that all sampled pixels from one SIF footprint remained together in one split. The polygon mean tree models used a date-grouped 80/20 train/test split stratified by month, measurement mode while the yield models used 2019–2022 region-year subset for training and 2024 subset as the test set. All these different methods of data partitioning were used to identify on a base level the ideal data partitioning technique.

Normalization was done for the predictor variables, and was calculated from the training partition only and then applied unchanged to validation and test data. A random seed of 42 was used for the data partitions and model training procedures. Raster channels were standardized separately, missing raster values were replaced by zero only after standardization as previously mentioned. Fractional footprint and aggregate masks were converted to floating-point values and normalized to sum to one before they were used to reduce a predicted SIF map to its supervised scalar value. For the CNN models, the SIF target were also standardized during optimization and then returned to the original units for evaluation using the following formula

$$SIF_{\text{standardized},i} = \frac{SIF_i - \mu_{\text{train}}}{\sigma_{\text{train}}}, \tag{5.9}$$

where μ_{train} and σ_{train} are the mean and standard deviation of the training targets. This transformation centred the targets near zero and gave them a standard deviation close to one, this helped support a more stable neural-network optimization. After inference, the predicted values were returned to the original SIF units using

$$\widehat{SIF}_i = \widehat{SIF}_{\text{standardized},i}\sigma_{\text{train}} + \mu_{\text{train}}. \tag{5.10}$$

It should be noted that the evaluation metrics such as RMSE and MAE were calculated in the original SIF units. The area-to-point MLP in contrast to the CNN models kept its target in the original SIF units. This was a design choice. Since CNN models contain more parameters than the MLP model, standardizing the SIF targets kept the loss and gradients on a consistent scale, and thus made optimization more stable and allowed the same learning settings to work across the various CNN model experiments. The CNN models and MLP model were optimized with AdamW using an initial learning rate of 10^{-3} and a weight decay of 10^{-4}. The CNN models using Sentinel-2 data processes four separate batches of eight chips (each chip is an image with multiple channels), and after each batch is done it calculates the gradients but doesn't update the model weights immediately. Only after all four batches are done are the gradients added together and finally the optimizer updates the weights. This method allows the model to be trained with a larger batch than what the GPU memory can permit, ie; the physical batch size is 8, while the effective batch size is 32. For a chip containing several footprints, the footprint losses were first averaged within the chip and then the chip losses were averaged across the batch. This helped prevent the chips containing more footprints from influencing the model training more greatly than chips containing less number of footprints, ie; avoid footprint count in a chip from influencing the model training performance. Huber loss was used for the models using combined SIF target (`target_modis_sif`) and for area-to-point, while mean squared error was used for the 740nm baseline model. Huber loss was used to stop extreme outliers from affecting the model training greatly. The Random Forest and XGBoost models were trained with fixed configuration settings (min tree depth, min node leaves) so that the comparison focused on the input representation and model family rather than effective hyperparameter search.

All the deep learning network models final model checkpoint were selected using the lowest validation RMSE. Models using Sentinel-2 data used early stopping with patience of seven epochs and also reduced the learning rate of the model when validation RMSE stopped improving. The area-to-point MLP allowed training to up to 150 epochs with patience of 20. The MODIS based models used 30 epochs and retained the checkpoint with the lowest validation RMSE. For most of the CNN models, a linear calibration of observed SIF versus predicted SIF was fitted on the validation predictions only. This was then applied unchanged to the test predictions. Both raw and calibrated metrics were kept so that any improvement caused by this post training adjustment remained visible, see FILLREF to see which models chose the calibrated results. The external native footprint inference used the saved Spatial aggregate (4km) model, its training normalization statistics, and its validation derived calibration to centred 4 km windows from two previously unseen MGRS tiles. These 4km windows had 20m SIF maps (4km cause each input chip is 200 pixels by 200 pixels, each pixel 20m resolution) which were averaged over the individual OCO-2 footprint masks and compared with their observed footprint SIF values. Finally the winter wheat yield model used a fixed Random Forest configuration for all crop pure and raw SIF predictor subsets, so that the 2024 test data comparison between the various yield models reflected more of the input data quality rather than any hyperparameter tuning efforts. A more concise view of the experimental settings can be seen in Table 5.1.

5.6 Experimental Environment and Reproducibility

For this study, all data acquisition, preprocessing, spatial matching, raster extraction and model ready data preparation were performed on a local workstation with a 20-core CPU, 32 GB RAM and RTX5070 GPU. For the python processing scripts, Python 3.11.9 in the Positron development environment was used. Python libraries like `NumPy` and `pandas` were used for tabular operations, `Rasterio`, `GeoPandas`, `Shapely` and `pyproj` were used for raster, vector, geometry and coordinate reference system (CRS) operations. For the R processing scripts, R 4.5.2 in the RStudio development environment was used. R libraries, `terra` and `sf` provided the main raster and vector functions, the `future`, `furrr`, and `tidyverse` packages provided the necessary parallel and tabular workflow functions. Processing operations that were computationally expensive were made easier to compute by grouping observations by date, product (or tile), cropping rasters before extraction and processing independent groups in parallel where possible. Since the local workstation had 32GB RAM and each Sentinel-2 reflectance band size can vary from 53MB to 230MB, crop type rasters around 450MB, without doing any kind of grouping, cropping the memory costs can easily baloon to over 32GB RAM. Eventhough some serial workflows for the processing could be done in memory with the ungrouped data, to make use of parallel computing, a smaller sized data needed to be made such that it could be fed to each core of the parallel process. All Neural Network training and evaluation were conducted in Kaggle environments which had two NVIDIA T4 GPUs, a four-core Intel Xeon CPU allocation, and 30 GB of RAM, the provided GPUs were better the local workstation's RTX5070 for model training. The Kaggle models used PyTorch 2.10.0 with CUDA 12.8. Training experiment settings like random seeds, data-partition definitions, normalization statistics, chip metadata, prediction tables, and evaluation outputs were stored with the corresponding jupyter notebooks so that each experiment could be redone without regenerating new partitions, normalization statistics for future steps. The best model; Spatial aggregate (4km) model weights, checkpoint, normalization statistics, and channel-order metadata are provided in the following google drive link FILLREF.

5.7 Evaluation Metrics Application

This section will describe the performance metrics that were used to evaluate the models and also Huber loss function that was used for the main CNN models training loss function. The residual as shown in FILLREF twice can be described as $e_i = \widehat{y}_i - y_i$ where y_i denotes the observed SIF or yield value for test sample i and $\widehat{y}_i$ denotes the predicted SIF value or yield value. Residual shows the prediction error for one observation. Positive residual indicates the model over predicted likewise negative value signifies under prediction. Similarly bias error indicates the mean of all residuals and therefore positive and negative values can use the same interpretation. An important point positive and negative residuals can cancel each other, so even if individual errors are large, bias can be near zero. Root mean squared error (RMSE), mean absolute error (MAE), and mean bias error described the predictive error while the coefficient of determination (R^2) measured the fraction of observed variation explained by the predictions. An R^2 value near one indicates that the model explains a large proportion of the variation in the observed values, while a value below zero means that the predictions perform worse than using the observed mean as a constant estimate ie; as a mean baseline model. Since RMSE depends on the spread of the target, normalized RMSE (NRMSE) was also used when comparing targets at different spatial supports, specifically for the Spatial aggregate model (4km) and multi footprint Sentinel-2 model (MultiSIF) comparison. These model metrics formulas can be seen in Section 2.3. Huber loss was used the optimization for the main CNN models and the area-to-point MLP model, with the exact transition parameter varying experiment to experiment as shown in Table 5.1. Huber loss behaves quadratically for small residuals but behaves linearly (linear penalty) for large residuals. This makes model fitting less sensitive to unusually large SIF errors than the mean squared error. Huber loss was used to update the model parameters during training (reduces this loss during training) whereas the metrics RMSE, MAE, bias, R^2 where used to report predictive performance metrics on the validation and test data sets.

Model performance was also further examined using the following metrics: slope and intercept of the least-squares regression of predicted values on observation values, Pearson's correlation coefficient, and distribution of residuals as seen in FILLREF. An ideal model would have a slope near 1, an intercept near 0, a correlation near 1, and residuals narrowly centred around 0. Pearson correlation measures how strongly predicted and observed values change together. The regression slope is noted by b_1, intercept of the regression line b_0, Pearson correlation r as seen in Section 2.3. An ideal model has $b_0 = 0$, $b_1 = 1$, zero bias, and small residual spread. Residual stability was also described by its standard deviation and interquartile range ($Q_{0.75}(e) - Q_{0.25}(e)$) and reported for Spatial aggregate (4km) and MultiSIF Sentinel-2 model, this described the spread and stability of prediction errors. For the Spatial aggregate (4km) model, predictor importance was estimated by independently permuting each validation channel (each channel of the image chip in the validation set) and checking the increase in RMSE $I_j = \mathrm{RMSE}_{\mathrm{perm}(j)} - \mathrm{RMSE}_{\mathrm{base}}$, with larger positive values indicating greater predictive importance. The core performance metrics as shown above were calculated for SIF value bins, footprint counts, measurement modes, months, years, MGRS tiles, states, and plant hardiness zones to identify differences in model performance among data groups. Relative RMSE reduction of crop-pure SIF against a raw-SIF baseline was calculated as $100(\mathrm{RMSE}_{\mathrm{raw}} - \mathrm{RMSE}_{\mathrm{crop}})/\mathrm{RMSE}_{\mathrm{raw}}$ and was used for comparison of the different yield models. Finally, given the small number of samples in the held-out 2024 test set, paired bootstrap resampling by region was used to quantify uncertainty in the RMSE difference between the crop-pure and raw-SIF yield models. In each bootstrap repetition, the same test regions were sampled with replacement for both models. This allowed for paired comparison. The 2.5th and 97.5th percentiles of the relative RMSE reduction distribution were used to report the 95% confidence interval.

Table 5.1: Training and partition settings used in the modelling experiments.

Experiment	**Split Unit**	**Loss**	**Optimizer**	**Batch**	**Epochs/trees**	**LR**	**ES**	**Calibration**
4 km aggregate U-Net	Component 80/10/10	Huber (1.0)	AdamW	8×4	30	10^{-3}	7	Yes
Sentinel-2 multi-SIF U-Net	Date 80/10/10	Huber (1.0)	AdamW	8×4	30	10^{-3}	7	Yes
MODIS multi-SIF U-Net	Year 2019–22/23/24	Huber (0.5)	AdamW	128	15	10^{-3}	-	Yes
MODIS single-SIF U-Net	Stratified 70/15/15	Huber (0.5)	AdamW	256	30	10^{-3}	-	Yes
MODIS 740 nm baseline	Stratified 70/15/15	MSE	AdamW	512	30	10^{-3}	-	No
Area-to-point MLP	Footprint 70/15/15	Huber (0.1)	AdamW	-	150	10^{-3}	20	No
Polygon means: XGBoost	Date 80/20	Squared error	XGBoost	-	800	0.03	-	No
Polygon means: Random Forest	Date 80/20	Squared error	RF	-	700	-	-	No
Winter-wheat yield RF	Year 2019-22/24	Squared error	RF	-	500	-	-	No

6 Results and Discussions

6.1 Experimental Overview

This section will give an overview of the experiments done in this study and expected result. The first stage evaluated how accurately OCO-2 SIF could be predicted from satellite derived spectral, biophysical, radiation, seasonal and crop composition under several model, input data representations and target representations. The evaluated models include polygon mean Random Forest and XGBoost regression, area-to-point multilayer perceptron, single footprint U-Net (MODIS predictors), multi footprint U-Net (MODIS and Sentinel-2 predictors) and the principal Spatial aggregate (4km) U-Net. The spatial supports (data representation) ranged from tabular footprint level summaries to spatially structured image chips (with multiple channels) and therefore tested both model architecture & choice and how the predictor information was represented. The initial baseline models used target SIF at 740nm before moving on to the uncertainty corrected combined SIF target derived from SIF at 757nm and 771nm as the later was found to increase the signal-to-noise ratio and thus make any SIF predictions more reliable. Training, validation, and test data were partitioned according to each dataset's temporal and spatial structure, complete dates, footprints, or connected date-product components kept together where used as split units, as described in Table 5.1. The model performance was assessed using RMSE, MAE, bias, R^2, regression slope, residual behaviour, data subgroup metrics and observed-versus-predicted SIF comparisons.

The results answer more general SIF prediction performance to more specific questions about data representation, model generalizability & stability, spatial variation and agricultural utility. The best model; Spatial aggregate (4km) was internally evaluated on held out chips (image with channels) and then externally evaluated on two previously unseen MGRS tiles on 4km windows (to match with input data extent) which produced 20m SIF maps which were then aggregated for each OCO-2 footprint and compared to original SIF observation. This external testing allowed to confirm if 20m SIF predictions aggregated to OCO-2 footprint resolution (1km^2) was viable. If confirmed, it would indicate the model is able to at least make SIF predictions at 1km^2 resolution and thus tackle the task of spatially and temporally completing a SIF map at OCO-2 SIF resolution. Comparisons between tree based, MLP and CNN models show how tabular summaries and spatial predictor grids affected accuracy of the SIF predictions. The effect of spatial aggregation on SIF prediction and retrieval-noise reduction is examined by comparing native footprint (MultiSIF Sentinel-2 model) and 4km targets (Spatial aggregate 4km model), their observed vs predicted SIF values variability, residual distributions, and performance across different footprint counts. Model behaviour is also evaluated across data subgroups for the following predictor variables: months, measurement modes, years and plant-hardiness zones which then helps identify conditions associated with stronger or weaker predictions. Finally the application of high resolution SIF predictions in winter wheat yield models were evaluated. Crop pure enhanced SIF was compared with raw mixed-footprint OCO-2 SIF at varying winter wheat coverages for select Bavarian NUTS-3 regions, testing was done on the held out 2024 subset. These results would then be compared with existing SIF enhancement approaches.

Table 6.1: Calibrated model performance by observed SIF interval for the combined nine-tile dataset, the held-out nine-tile test set, and the external native-footprint evaluation over two additional tiles.

A. Combined training, validation and test data: nine tiles								
SIF interval	n	**Obs. mean**	**Pred. mean**	**RMSE**	**MAE**	**Bias**	R^2	**Pearson** r
$[-0.25, 0)$	66	−0.0246	0.0851	0.1264	0.1097	0.1097	−32.616	–
$[0, 0.25)$	2,920	0.1373	0.1560	0.0645	0.0466	0.0187	−0.028	–
$[0.25, 0.50)$	3,846	0.3721	0.3631	0.0662	0.0520	−0.0090	0.091	–
$[0.50, 0.75)$	1,076	0.5712	0.4718	0.1187	0.1008	−0.0995	−3.380	–
$[0.75, 1.00)$	12	0.7873	0.5149	0.2756	0.2724	−0.2724	−56.094	–
B. Held-out test data: nine tiles								
SIF interval	n	**Obs. mean**	**Pred. mean**	**RMSE**	**MAE**	**Bias**	R^2	**Pearson** r
$[0, 0.25)$	274	0.1509	0.1761	0.0658	0.0437	0.0251	−0.247	–
$[0.25, 0.50)$	545	0.3636	0.3455	0.0605	0.0479	−0.0181	0.177	–
$[0.50, 0.75)$	112	0.5643	0.4667	0.1119	0.0983	−0.0976	−3.810	–
C. External native-footprint evaluation: two additional tiles								
SIF interval	n	**Obs. mean**	**Pred. mean**	**RMSE**	**MAE**	**Bias**	R^2	**Pearson** r
$[-0.5, -0.25)$	4	−0.2689	0.3181	0.5876	0.5870	0.5870	−3884.368	0.999
$[-0.25, 0)$	104	−0.0625	0.1816	0.2798	0.2441	0.2441	−25.084	−0.055
$[0, 0.25)$	652	0.1413	0.2249	0.1534	0.1174	0.0836	−4.232	0.274
$[0.25, 0.50)$	827	0.3707	0.3617	0.1104	0.0893	−0.0091	−1.375	0.307
$[0.50, 0.75)$	350	0.5942	0.4319	0.1864	0.1634	−0.1623	−6.672	0.184
$[0.75, 1.00)$	59	0.8195	0.4491	0.3832	0.3704	−0.3704	−44.830	−0.141
$[1.00, 1.25)$	4	1.1151	0.3941	0.7233	0.7210	−0.7210	−213.446	0.324

Table 6.2: Calibrated fitted performance across measurement mode, temporal alignment, month, and observation-year subgroups.

A. Measurement mode								
Mode	**Obs. min**	**Obs. max**	**Obs. mean**	**Pred. mean**	**RMSE**	**MAE**	**Bias**	R^2
Mode 0	0.0306	0.7200	0.3105	0.3030	0.0659	0.0492	−0.0075	0.8090
Mode 1	0.0131	0.6790	0.3412	0.3181	0.0746	0.0566	−0.0231	0.6885
B. Date alignment								
Alignment	**Obs. min**	**Obs. max**	**Obs. mean**	**Pred. mean**	**RMSE**	**MAE**	**Bias**	R^2
Within range	0.0131	0.7200	0.3379	0.3219	0.0712	0.0539	−0.0160	0.7421
Outside range	0.0570	0.4855	0.1820	0.1794	0.0569	0.0393	−0.0026	0.6428
C. Observation month								
Month	**Obs. min**	**Obs. max**	**Obs. mean**	**Pred. mean**	**RMSE**	**MAE**	**Bias**	R^2
February (2)	0.0285	0.1449	0.0869	0.0861	0.0302	0.0236	−0.0008	−0.2443
March (3)	0.0452	0.3008	0.1267	0.1172	0.0343	0.0263	−0.0095	0.4035
April (4)	0.0131	0.5578	0.2999	0.2890	0.0582	0.0453	−0.0109	0.5123
May (5)	0.0828	0.7200	0.4493	0.4193	0.0876	0.0712	−0.0301	0.4355
June (6)	0.2119	0.6527	0.4200	0.4035	0.1025	0.0753	−0.0165	−0.2676
July (7)	0.0441	0.5177	0.3447	0.3463	0.0816	0.0609	0.0016	0.2947
D. SIF observation year								
Year	**Obs. min**	**Obs. max**	**Obs. mean**	**Pred. mean**	**RMSE**	**MAE**	**Bias**	R^2
2019	0.0131	0.6527	0.3117	0.3021	0.0654	0.0493	−0.0095	0.4809
2020	0.0452	0.3008	0.1267	0.1172	0.0343	0.0263	−0.0095	0.4035
2022	0.0828	0.7200	0.4482	0.4190	0.0890	0.0721	−0.0293	0.4202
2023	0.2315	0.5481	0.4099	0.4069	0.0729	0.0579	−0.0030	0.0938
2024	0.0285	0.1449	0.0869	0.0861	0.0302	0.0236	−0.0008	−0.2443

6.2 Overall SIF Prediction Performance

The primary model for this study; Spatial aggregate (4km) U-Net was evaluated first on held out 4km grid cells from the nine MGRS tiles (internal dataset for training, validation, testing) and later evaluated on native OCO-2 footprints from two additional MGRS tiles. This testing process allowed to see how the performance changed when the spatial support was changed from 4km grid to the noisier and finer native footprint support, remember the spatial aggregation was done primary to reduce the retrieval noise during data acquisition. The calibrated model reported RMSE of 0.0701 and R^2 of 0.761 for 931 grid cell targets for the held out 4km test set as shown in Figure **??**. The predictions regression on observations had a slope of 0.726 and an intercept of 0.074. Therefore the model was able to explain a substantial part of the variation in the aggregated SIF target. The slope below one shows that its predicted range was narrower than the observed range, ie; low SIF values were overpredicted while the largest values were generally underestimated. One reason could be the lower sample size of these values as seen in the combined fitted SIF bin performance table FILLREF. In light of this, the strong spatial aggregate result should be interpreted as accurate prediction of the noise reduced 4km SIF means rather than a direct validation for every 20m output SIF pixel. To verify if this strong performance can be carried out to a much lower resolution, it was validated with the lowest resolution real SIF data (OCO-2 SIF footprints).

The external native footprint experiment applied the same trained model and validation derived post training calibration to 2,000 individual OCO-2 footprints in the two new MGRS tiles that were not seen during model training. For the 2,000 footprints, the calibrated RMSE was 0.1701, R^2 was 0.373, and regression slope was 0.400 (Figure **??**). The mean bias was only -0.0034, but this small value could be the result of positive and negative residual values cancelling each other out as previously discussed. The external test plot shows strong range compression, with low and negative observations predicted too high and high observations predicted too low. Given that our SIF range seen during model training was around $[0, 0.75)$, 1,829 external footprints evaluation with SIF range $[0, 0.75)$ reduced RMSE to 0.1424 and increased slope to 0.472 while R^2 remained 0.349 (Figure **??**). The lower RMSE is expected since the reduced external SIF range matches closely to the training SIF range. No change in R^2 can attributed to the native SIF samples being extremely noisy as can be seen in (Figure **??**), but high density of points can still be seen closer to the diagnol.

The interval-specific results in Table 6.1 can clarify this behaviour. In the held-out nine tile test set, RMSE was 0.0658 for SIF in $[0, 0.25)$ and 0.0605 for $[0.25, 0.50)$, but increased to 0.1119 for $[0.50, 0.75)$. The corresponding bias changed from $+0.0251$ in the lowest interval to -0.0976 in the highest interval, thereby confirming the tendency of the model to pull predictions towards the centre of the target distribution. This pulling effect was stronger at native footprint scale where RMSE decreased from 0.1534 in $[0.0, 0.25)$ to 0.1104 in $[0.25, 0.50)$ and increased to 0.1864 in $[0.50, 0.75)$ and 0.3832 in $[0.75, 1.00)$. The very small sample counts in the extreme intervals mean that their individual statistics are uncertain, but the overall direction of the errors is consistent with the observed-versus-predicted plots. We can see that model was able to predict well for native OCO-2 SIF resolution for the SIF range $[0.0, 0.75)$. It should be noted that negative within-bin R^2 values should not be read as a contradiction of the overall test R^2 because each narrow SIF interval contains much less target variation, so even moderate absolute errors can perform worse than that's interval mean predictor. Spatial aggregation largely removed negative SIF samples and therefore it cannot be expected for the model to predict them in the native external set.

The performance also varied across acquisition date, temporal subgroups, measurement modes (Table 6.2). Measurement mode 0 (Nadir) subgroups had an RMSE of 0.0659 and an R^2 of 0.809 versus 0.0746 and 0.689 for mode 1 (Glint), this slight variation is expected as Nadir will capture the signals more strongly than Glint FILLREF. SIF observations whose acquisition dates were within (ingrange) the temporal range of their Sentinel-2 products had a higher R^2 than observations outside (outrange) that range (0.742 versus 0.643). The date_align RMSE values were lower for outrange samples because its observed SIF range was much smaller than inrange samples. For the months, we can see the RMSE increasing monthwise till June, this corresponds to the SIF range also being more variable as the months proceeded. Year specific results should be taken with caution as the available months and sample counts differed among years, therefore it does not form a controlled test for generalization among different years.

Feature importance for this model was acquired by permutation importance of the validation set. Figure **??** shows that FAPAR caused the largest increase in validation RMSE when permuted, followed by the seasonal cosine channel and NIRv. Since SIF $\approx$ APAR $\times$ Φ_F $\times$ f_{esc} = PAR $\times$ fAPAR $\times$ Φ_F $\times$ f_{esc}, SIF and GPP share dependence on absorbed radiation, it makes sense that FAPAR was the most important variable, likewise FILLREF showed NIRv to be important in their SIF enhancement model and the same effect can be seen here too. APAR, NDVI and EVI were the next influential predictor variables and PAR alone, most crop faction channels had relatively small RMSE improvement effects. The higher importance of APAR than PAR shows that the absorbed radiation supplied more useful information to the model than the incoming radiation. Among the crop channels, the active crop channel contributed more to the RMSE improvement than the individual crop fractions, signifying seasonal representation of crop composition data provided more useful information to the model. It should be noted that these values are conditional permutation results rather than causal effects. To reiterate, it shows how much the trained model relies on each predictor given all the other predictors are available, it does not prove that changing the predictor physically causes the SIF to change. Also another important point, correlated predictors can substitute for one another, therefore any near-zero or negative importance can indicate redundacy of some predictors rather than physical irrelevance.

The spatial residual map for the internal test samples as seen in Figure **??** shows the spatial distribution of over and underestimation of SIF values. Positive and negative residuals occurred across the nine tiles showing that the overall error was composed of spatially varying local outcomes rather than one uniform outcome. This map is useful for identifying locations where differences in land cover, predictor quality, acquisition timing or distribution of the OCO-2 footprints may have affected the prediction of SIF values. It should be noted that this residual map is only for the nine tiles internal testing and doesn't include external native footprint residuals. The terminal colours on the map represent both the clipping threshold and any extreme residuals as the colour scale was clipped symmetrically at 98 percentile (($\pm$0.180)). The final results for all these figures and tables show that high resolution spectral, biophysical, radiation, seasonal and crop information predicted noise reduced 4km SIF target with good accuracy. This model was able to carry out its performance for the native external set for the SIF range $[0.0, 0.75)$ particularly. Since the scope of this study is only for Germany, acquiring noise 4km SIF samples outside this range may not be possible or may not yield a sizeable sample count for the model to train on $[0.75, 1.5)$ SIF range samples. Finally the predicted 20m SIF maps should be described as spatial downscaling limited to the fine resolution predictors and aggregate OCO-2 supervision, it cannot be validated as 20m SIF values as no real SIF dataset exists at 20m resolution.

6.3 Effects of Model Architecture and Input Representation

Model architecture comparison covered the tree based regressions, area-to-point multilayer perceptron and several U-Net configurations, each using either polygon level tabular predictors or spatial raster chips (images with channels). Table 6.3 shows that the Spatial aggregate (4km) U-Net achieved the strongest internal result, with an RMSE of 0.0701 and an R^2 of 0.761. When tested on an external native OCO-2 footprint resolution, it was able to achieve an RMSE from 0.1534 to 0.1864 for the SIF range $[0.0, 0.75)$, but it was not able to carry out the strong performance outside this range as the internal training dataset SIF range was limited to around $[-0.25, 1)$ with only about 78 samples for the extreme ranges $[-0.25, 0)$ and $[0.75, 0)$. For the following native footprint experiments (single, multi SIF footprints, mean aggregated predictions), the multi footprint Sentinel-2 model obtained an RMSE of 0.1603 and R^2 of 0.418. The MODIS single-footprint model was similarly competitive, with an RMSE of 0.1656 and an R^2 of 0.457. This indicated that both 20m and 250m base predictor resolutions were suitable for SIF predictions. Polygon-means XGBoost and Random Forest produced RMSE values of 0.1650 and 0.1689, with R^2 values of 0.453 and 0.426, respectively. This also showed reducing the predictor values to one single mean value over the target support was competitive with the CNN models that could see internal spatial configurations. The small differences between these native footprint experiments indicate that retaining a spatial structure for the predictors did not automatically improve footprint level predictive accuracy over polygon mean predictors. The MODIS baseline model and the area-to-point MLP used SIF at 740nm as the target variable, they reported much higher RMSE values of 0.2693 and 0.2503 than the models that uses the combined SIF target derived from SIF at 757nm and 771nm. This indicated that the combined SIF target was able to increase the signal-to-noise ratio compared to the SIF at 740nm target. It should be noted that the results compare complete modelling systems that differ in target definition, predictor source, spatial support, calibration, data partition and sample size, therefore it cannot theoretically provide a controlled test of model architecture alone.

The observed-versus-predicted plots for all the native footprint models shown in Figure **??** show a compression of the predicted SIF range toward its centre. We can see the spatial aggregate (4km) model is able to keep the predictions more closer to the diagonal line than the native footprint models ie; the distribution is more closer to the line indicating the noise reduction. The fitted slopes were approximately 0.439 for the multi footprint Sentinel-2 model, 0.444 for Random Forest, and 0.465 for XGBoost, while for the spatial aggregate (4km) model it had a much higher slope of 0.726. We can also see that the spatial aggregate model can predict SIF values better in the $[0.0, 0.75)$ range than the native footprint models, extreme SIF ranges remain a challenge for all the models. Likewise the residual distributions in Figure **??** support the same interpretation: the Spatial aggregate (4km) model had a much narrower residual distribution, while models evaluated against individual footprints showed broader tails. A large part of this difference can be attributed to the noise reduction produced by aggregating several close OCO-2 observations rather than the model architectures. The SIF prediction maps along with equivalent NDVI map in Figure **??** demonstrate an important representation advantage of the image based models as they are able to reflect patterns much more complex because their output preserved field scale spatial patterns and surrounding landscape spatial context. XGBoost and RandomForest (not shown) was also applied to form a 20m SIF predicted map since each predictor variable had a 20m resolution, but it should be noted that each SIF pixel was predicted independently and any visual spatial continuity in the predicted map is from the predictor raster structure rather than any learned neighbourhood relationships. These predicted SIF maps are spatial dowscaling products rather than direct pixel level validated maps (since no true 20m SIf data exists). To summarize the results, target preparation and spatial support (predictor representation) had a larger influence on the accuracy of prediction rather than the choice of the model architectures (tree based, MLP, CNN).

6.4 Spatial Aggregation, Stability and Generalization

Spatial aggregation of SIF observations (nearest samples averaged to one value for one grid cell) greatly reduced the variability of the observed targets and also model errors. The standard deviation (SD) of the observed SIF decreased from 0.210 for multi SIF (multiple sif footprints in one window but treated as individual footprints. We can also call these native footprint) to 0.144 for 4km spatial aggregates, the interquartile range for the SIF value also decreased from 0.291 to 0.203 shown in Table 6.5, signifying SIF range compression as a result of spatial grouping. The residual standard deviation dropped from 0.159 to 0.069 (native footprint to spatial aggregate) and the normalized RMSE improved from 0.763 to 0.488. NRMSE was used for strict comparison purposes to account for the fact that observed SIF SD fell from 0.210 to 0.144, individual model effectiveness will use RMSE. So after taking care of this SD change, the RMSE dropped thereby showing that the spatial aggregate 4km model still performed better even after accounting for the smaller variation in the aggregated SIF values. Figure **??** shows a tighter observed vs predicted relationship for the aggregated model, showing less scatter around the fitted regression line and a slope closer to one. Footprint count analysis as seen in Figure **??** shows additional evidence that spatial aggregation reduced SIF retrieval noise, where the mean target standard error declined from approximately 0.060 for chips (4km windows) containing five or six footprints to about 0.040 for chips containing at least thirteen. Likewise RMSE decreased from about 0.080 for five or six footprints to about 0.062 for eleven or twelve footprints. We can see a slight increase in RMSE for thirteen or more footprints, this could indicate that adding more observations to 4km window may not necessarily guarantee improvement in prediction as high footprint count chips may represent neighbouring cells that are different from each other. Therefore increased spatial density of OCO-2 footprints in a 4km window only slightly reduces the uncertainty of the reconstructed SIF product. Spatial aggregation improved target stability and predictive accuracy but it also narrowed the target range from approximately -0.25-1.25 at native support to 0.0-0.72 at 4km support and removed most negative and very high SIF observations.

Table 6.1 and Table 6.4 provides the SIF interval level results. For the held out 4km test set (internal), RMSE was 0.066, 0.061, and 0.112 in the $[0, 0.25)$, $[0.25, 0.50)$, and $[0.50, 0.75)$ SIF intervals showing strong performance whereas for the native footprint model, higher RMSE values of of 0.136, 0.111, and 0.201 was reported for the same SIF intervals. The Spatial aggregate (4km) model was tested out on an external native footprint test set and showed RMSE values 0.153, 0.110, and 0.186 which was closer to the native multi footprint model. One of the major reason for that is the native footprint is quite noisy inherently, but visual inspection of the external observed vs predicted plot in (Figure **??**) shows there is a high density of points close to the diagonal line. The aggregate model's external errors were large for observations outside the aggregate training range, reaching an RMSE of 0.280 for $[-0.25, 0)$, 0.383 for $[0.75, 1.00)$, and 0.723 for $[1.00, 1.25)$ thereby showing that learned relationships in the $[0.0, 0.75)$ range cannot be carried over to the extreme SIF intervals. Violin plots as shown in Figure **??** illustrates the effect of spatial aggregation, where the greater width of the violin indicates a higher concentration of observations. The 4km model's residual distribution is more concentrated around zero compared to the native footprint residuals which have a wider spread and longer tails. This indicates that the aggregation reduced target variability and increased prediction stability.

Table 6.3: Comparison of the calibration status, SIF target, sample size, and fitted performance of the evaluated RQ2 model configurations.

Model	Calibration	SIF target	n	RMSE	MAE	Bias	R^2	Pearson r
Spatial Aggregate 4 km (SA4km)	Calibrated	MODIS SIF	931	0.0701	0.0527	−0.0149	0.761	–
SA4km OCO-2 native inference	Calibrated	MODIS SIF	2,000	0.1701	0.1300	−0.0034	0.373	0.612
SA4km OCO-2 native, SIF $[0, 0.75)$	Calibrated	MODIS SIF	1,829	0.1434	0.1135	−0.0054	0.349	0.613
MultiSIF-Sen2	Calibrated	MODIS SIF	6,411	0.1603	0.1223	−0.0166	0.418	–
MODIS 24 × 24	Calibrated	MODIS SIF	2,187	0.1679	0.1293	0.0010	0.349	–
MODIS 16 × 16	Calibrated	MODIS SIF	22,149	0.1656	0.1255	0.0011	0.457	–
MODIS 16 × 16	Uncalibrated	740 nm SIF	26,905	0.2693	0.2107	0.0588	0.350	–
MLP area-to-point	Uncalibrated	740 nm SIF	5,291	0.2503	0.1912	−0.0053	0.415	0.646
PolyMeans: XGBoost	Uncalibrated	MODIS SIF	5,889	0.1650	0.1274	0.0125	0.453	0.675
PolyMeans: Random Forest	Uncalibrated	MODIS SIF	5,889	0.1689	0.1311	0.0146	0.426	0.656

Table 6.4: Calibrated MultiSIF test performance across observed SIF intervals.

SIF interval	n	Obs. min	Obs. max	Obs. mean	Pred. mean	RMSE	MAE	Bias	R^2
$[-0.50, -0.25)$	1	−0.2513	−0.2513	−0.2513	0.4294	0.6808	0.6808	0.6808	–
$[-0.25, 0)$	306	−0.2329	−0.0002	−0.0555	0.1689	0.2650	0.2244	0.2244	−25.959
$[0, 0.25)$	2,240	0.0001	0.2499	0.1411	0.2064	0.1360	0.1015	0.0653	−3.013
$[0.25, 0.50)$	2,606	0.2502	0.4997	0.3664	0.3394	0.1105	0.0894	−0.0270	−1.546
$[0.50, 0.75)$	1,063	0.5001	0.7495	0.5943	0.4227	0.2008	0.1743	−0.1716	−7.940
$[0.75, 1.00)$	186	0.7511	0.9889	0.8269	0.4832	0.3567	0.3437	−0.3437	−34.460
$[1.00, 1.25]$	9	1.0281	1.2161	1.0891	0.5101	0.5854	0.5790	−0.5790	−117.500

Table 6.5: Comparison of the spatially aggregated and multi-footprint Sentinel-2 U-Net models. Continuous values are rounded to three decimal places.

Metric	Spatial U-Net	Multi-footprint U-Net
Supervision	Spatially aggregated SIF	Individual SIF footprints within multi-footprint chips
Target support	4 km fixed-grid cell	Native OCO-2 footprint
Predictor resolution	20 m	20 m
Training samples	6,060	4,983
Validation samples	929	553
Test samples	931	686
Test dates	21	18
Observed mean	0.325	0.320
Observed SD	0.144	0.210
Observed IQR	0.203	0.291
Observed minimum	0.013	−0.251
Observed maximum	0.720	1.216
RMSE	0.070	0.160
Normalized RMSE	0.488	0.763
MAE	0.053	0.122
Bias	−0.015	−0.017
R^2	0.761	0.418
Regression slope	0.726	0.439
Regression intercept	0.074	0.163
Residual SD	0.069	0.159

Since the aggregate internal, multi footprint, and aggregate external results were obtained from different spatial supports and test observations, we can only confirm that these comparisons describe general patterns rather than strict paired sample-by-sample patterns. In conclusion, aggregation provided useful SIF retrieval noise reduction, but since the aggregation caused the loss of extreme targets, the model couldn't learn these extreme targets. A larger scope of the dataset outside Germany is needed to get a larger sample size of SIF values in the range $[0.75, 1.25)$.

6.5 Agro-Climatic and Seasonal Variation in Model Performance

The combined fitted evaluation 6.6 uses the training, validation, and test predictions from the Spatial aggregate (4km) to examine inter plant hardiness zone performance given that the overall model performance has been established in the previous sections. Table 6.6 shows that zone 7b had the lowest RMSE at 0.0690, while zone 7a had the highest R^2 at 0.8681. SIF prediction performance was slightly weaker in zones 8a, 8b which had bigger sample sizes, they reported RMSE of 0.0772 and 0.0803 and R^2 0.7812 and 0.7553 respectively. Zone 9a had the highest RMSE of 0.0946 and largest bias of -0.0272, but this estimate was based on a small sample size of 58 observations obtained from 14 date tracks and two MGRS tiles. The choropleth maps in Figure **??** show the geographical distribution of the differences in RMSE values with respect to plant hardiness zones. It should be noted that the reported values represent broad zone level peformance rather than continuous local accuracy, ie; a single value for an individual plant hardiness zone. Figure **??** are the residual boxplots of each available plant hardiness zone and shows similar central residual distributions across the five zones, with residual medians generally below zero and occasional large positive and negative errors. Again, the smaller sample size for zone 9a makes it residual distribution less reliable compared to the other zones.

Table 6.6: Calibrated fitted performance by individual hardiness zone.

HZS	n	**Obs. min**	**Obs. max**	**Obs. mean**	**Pred. mean**	**RMSE**	**MAE**	**Bias**	R^2	**Dates**	**Tiles**
7a	151	−0.0496	0.7544	0.2477	0.2337	0.0703	0.0518	−0.0140	0.8681	30	1
7b	894	−0.0347	0.7462	0.3190	0.3038	0.0690	0.0517	−0.0153	0.8176	63	5
8a	4,458	−0.0937	0.8291	0.3066	0.2983	0.0772	0.0585	−0.0083	0.7812	201	9
8b	1,502	−0.0891	0.8783	0.3155	0.3021	0.0803	0.0608	−0.0134	0.7553	169	8
9a	58	−0.0040	0.7881	0.4114	0.3842	0.0946	0.0697	−0.0272	0.7752	14	2

Figure **??** shows the month versus plant hardiness zone RMSE performance plot, where the RMSE was lowest in February and March with values in 0.034-0.053 range. Errors increased for months April-July, with May and June reporting the highest RMSE values. RMSE reached 0.094 in zone 7a during May, 0.096 in zone 8b during June, and 0.090 in zone 8a during both June and July. There were some combination of zone-month values that count less than 20 or no data, these RMSE values were not reported as the sample size is too low to draw an meaningful conclusion. The increase in RMSE for the later months might be partly due to the expansion of SIF range values (wider SIF variation), changing crop conditions and differences in the spatial distributions rather than a purely climatic zone effect. Across the high sample size zones 7b, 8a, 8b, RMSE varied by only about 0.011, thereby suggesting that the model maintained reasonably stable peformance across different agro-climatic conditions. The month to month changes were larger than the differences between these zones, thereby indicating that the seasonal crop development and SIF variability mattered more on the prediction error than plant hardiness zone membership alone. Lastly the negative biases and residual medians indicate a small tendency for the model to under predict SIF across almost all zones.

6.6 Winter-Wheat Yield Prediction

The winter wheat yield experiments compared crop pure enhanced SIF, raw OCO-2 footprints with at least 5%, 10%, 30% wheat coverage and all available raw OCO-2 footprints in five MGRS tiles using a standard fixed configuration Random Forest. A fixed Random Forest configuration was used as the main purpose of this experiment was to see predicitive power of the individual data subsets rather than model configurations. It should also be noted that the 10% and 30% wheat coverage subsets were excluded from this result section as they were found to have very few reliable samples owing to the fact most of the SIF footprints had a low wheat farm area due to small farm plots and high mixed crop farms. Table 6.7 shows that the crop pure yield model achieved the lowest 2024 test RMSE at 0.9406 t/ha versus 1.0285 t/ha for the raw 5% wheat coverage model and 1.0874 t/ha for the all footprint model. These differences correspond to RMSE reductions of approximately 8.5% and 13.5% when crop pure SIF was used. The crop pure model also had the lowest MAE, bias, and residual standard deviation, but its normalized RMSE was above one and its R^2 was still negative at -0.1456 but higher than the raw SIF models. Figure **??** shows that all the three models compressed predictions toward the average yield. The crop pure model showed a clearer positive relationship with observed yield than either of the raw SIF models.

Figure **??** shows a mean crop pure RMSE improvement of about 0.084 t/ha over the 5%-wheat model, with an estimated probability of 0.879 that crop pure SIF produced the lower RMSE. It also shows the mean crop pure RMSE improvement of 0.143 t/ha over the all footprint model with a probability of 0.918. However it should be noted that, the 95% bootstrap intervals were $[-0.066, 0.213]$ and $[-0.052, 0.347]$ t/ha with both crossing zero, so the improvement was not statistically conclusive at the 95% level. Paired bootstrap resampling was used because the test set contained only 48 test samples (48 NUTS3 regions). It estimated how uncertain the RMSE difference was while ensuring that both models were repeatedly compared using the same sampled regions. The pair regional comparison as shown in Figure **??** shows an uneven spatial response. Crop pure SIF reduced absolute error in 27 of 48 regions relative to 5% model and 26 out of 48 regions relative to all footprint model. The takeaway is that the results provide preliminary evidence that crop pure enhanced SIF can improve winter wheat yield prediction compared to the raw SIF (mixed crop SIF signals), but the negative R^2 values and wide bootstrap intervals show that the present models are not yet sufficient for making claims regarding strong yield prediction claims. A negative R^2 means the model predicted the 2024 yields worse than a simple baseline that assigns every test region the mean observed 2024 yield. The reasons for this could be the very small test dataset and year to year distribution differences. Therefore the results can only support relative improvement from raw SIF to crop pure SIF, but cannot be treated as a reliable stand-alone yield prediction model. Additional observation years and more NUTS 3 regions would be needed to determine whether the crop pure SIF advantage remains stable in a larger yield dataset. Given the small number of yield model ready observations, extending this trained yield model to other c3 crops cannot be reliably done as mean aggregation to NUTS 3 scale will make the dataset much more sensitive to small changes. Therefore it is recommended to apply the same crop pure SIF extraction workflow separately for other c3 crops and generate the model ready dataset rather than applying a single yield model for all c3 crops in a given region.

Table 6.7: Comparison of winter-wheat yield-model performance using three monthly SIF aggregation strategies.

Model	n_{train}	n_{test}	RMSE	NRMSE	MAE	R^2	Bias	Slope	Intercept	Residual SD	Bootstrap RMSE interval
Crop-pure monthly SIF	197	48	0.9406	1.0591	0.7598	−0.1456	0.3507	0.1798	5.8570	0.8820	[0.7679, 1.1068]
Raw $\geq$ 5% wheat monthly SIF	197	48	1.0285	1.1581	0.7769	−0.3698	0.4395	0.0288	6.9595	0.9397	[0.7737, 1.2612]
Raw all-footprint monthly SIF	197	48	1.0874	1.2244	0.7911	−0.5311	0.4622	−0.0132	7.2645	0.9947	[0.8045, 1.3663]

6.7 Comparison with Existing Techniques

Out of the existing SIF enhancement in Table 3.1, SIFnet [GTK+22] and CNSIF [DXH+25] produces the highest resolution SIF estimates at 500m resolution. This thesis's best SIF enhancement model Spatial aggregate (4km) will be compared with these primarily as they use CNN models and similar predictors but their metrics are not directly equivalent because the SIF enhanced products use different SIF targets, spatial supports, timeframe and validation observations. For the internal test, the Spatial aggregate (4km) developed on February-July observations during 2019-2024 across nine German MGRS tiles obtained an RMSE of 0.0701 and a R^2 of 0.761 on 931 held out 4km (cell of 16km^2 area) aggregate targets. The external test on 2,000 native OCO-2 footprints from the same seasonal and yearly domain in two previously unseen German MGRS tiles produced an RMSE of 0.1434 and R^2 of 0.349, this validation was done at roughly 1km^2 resolution. SIFnet was trained for 16-day periods during 2018-2021 using five regions across Asia, Europe, souther Africa and South America with North America (CONUS) used as the held out internal test set. They were able to use such a large geographic scope because they used MODIS 500m resolution predictor variables which is much coarser than the 10m, 20m Sentinel-2 raster data. A single sinusoidal gridded (covering 1,112km by 1,112km area) MODIS data has around 2400 by 2400 pixels for one predictor whereas a single MGRS grid tile (covering 110km by 110km) data can have around 30.25m to 121m pixels, an extremely large difference. Its held out CONUS internal test at 0.05° scale was RMSE of 0.17 and R^2 of 0.92. SIFnet also externally validated their model on OCO-2 footprint samples in CONUS using a 0.005° support, obtaining a RMSE of 0.21 and R^2 of 0.57.

The thesis 4 km spatial model's external validation results are in line with those reported for SIFnet, showing that the model can learn a transferable relationship between the Sentinel-2 & secondary predictor variables and SIF. SIFnet has an important advantage because a spatially complete, TROPOMI SIF parent field is available to guide each downscaled estimate, whereas the thesis model has no parent SIF input and must perform OCO-2 gap filling and resolution enhancement together using auxiliary predictors alone which is a much more challenging problem. CNSIF conducted downscaling on GOSIF 0.05° (roughly 20km^2 area) target support on years 2003-2018, 2020-2022 and obtained R^2 of 0.856 and RMSE of 0.089 against a 2019 withheld GOSIF test set. Its external evaluation was on SIF data obtained through ChinaSpec ground tower sites in nine select local regions in China covering the years 2017-2021 and reported a pooled R^2 of 0.581 and RMSE of 0.152. This is less comparable with the thesis external evaluation compared to SIFnet, but still provides a reasonable range of expected performance metrics. A direct comparison of the yield models with existing crop yield studies cannot be justified because their geographic domains, predictors, crops, and training sample sizes differ. The supported comparison is instead internal for the thesis yield model, where crop-pure enhanced SIF reduced 2024 Bavarian NUTS-3 RMSE by 8.5% relative to the 5% wheat coverage mixed footprint signal and by 13.5% relative to all raw OCO-2 footprints, providing preliminary evidence that isolating crop-specific SIF can be more useful than retaining mixed signals.

7 Conclusion

7.1 Summary

Prior research for SIF enhancement has shown that sparse and coarse satellite SIF data can be reconstructed, spatially enhanced (increased resolution) using optical, radiation, meteorological and land cover variables [LX19, ZJA+18, SWG+22]. Reconstruction methods mainly on OCO-2 and OCO-3 data give spatial and temporal completeness, but any fine scale values produced are mainly inferred from predictor relationships with SIF and do not usually have a requirement (passed as another predictor) to preserve the coarse SIF observation magnitude from the same time. In contrast TROPOMI based downscaling methods [HJM+24, CLS+25, ZZZ23] can retain more of the original measured SIF signal by giving its magnitude bounds as a predictor for downscaling the coarse signal, ie; the model will know what is the general range of SIF at the particular pixel it is supposed to downscale thereby reducing the range for prediction, OCO-2 and OCO-3 cannot provide such coarse signal for every point on Earth. These TROPOMI based methods retain the original signal by bias correction, redistribution and physical observation constraints. Reconstruction and Downscaling methods that used a convolutional representation of data showed that spatial predictor grids can represent neighbourhood patterns and mixed land cover patterns that would generally be unavailable to tabular models containing mean aggregated predictor values[GTK+22, DXH+25]. However the finer SIF output does not guarantee that the estimated subpixel SIF pattern represents true physiological variation because many of the predictors can represent more about the canopy structure rather than short term photosynthetic regulation, ie; canopy structures changes in a much more slower timeframe than photosynthesis process. This limitation is also more difficult to address with OCO-2 than with TROPOMI because the narrow and widely separated OCO-2 tracks leave large areas without a parent SIF value that could give magnitude bounds to the redistribution method [YWC+19]. Also the predictors can vary slightly in terms of data acquisition and therefore the final fine scale output represents more about the average subpixel SIF distribution than a instantaneous distribution.

The previous highest resolution SIF products also remain at approximately 500m resolution or coarser, which is insufficient for isolating crop pure SIF signals for many small and irregularly shaped agricultural fields especially in a German setting. Model generalizability and fine scale validation is also a common limitation in the papers shown in Table 3.1, as many enhanced SIF products are assessed against held out supervisory observations (same scale, same region, same satellite) or indirect proxies such as GPP and yield rather than independent SIF measured at the output fine resolution. Though the latter is more due to the lack of availability of SIF observations higher than 1km^2 resolution, though newer satellite observations like FLEX ESA [DMDB+16, CTP+17] will soon observe 300m resolution SIF data which can then help validate these methods more reliably. Crop yield and productivity studies show that SIF is the most important remote sensing data for the purpose, accurate estimations depends on the crop type, observation timing, geographic scale and inclusion of adition secondary predictors like weather, soil moisture and other environmental predictors [PGZ+20, SGG+18, QLH+22]. The existing SIF resolution limitation and its application to German crop yield prediction motivated the present study to test OCO-2 SIF enhancement with Sentinel-2 predictors at 20m resolution and examining whether the resulting crop pure signal benefits winter wheat yield prediction.

The experimental workflow for this thesis consisted of two connected stages: SIF enhancement followed by regional winter wheat yield prediction. The first stage used quality controlled OCO-2 observations from February to July during 2019-2024 across nine MGRS tiles in Germany, with two further tiles reserved for external evaluation (model generalizability). SIF quality processing involved combining 757nm and 771nm observations, along with their corresponding noise retrieval uncertainity corrections and checks applied producing a final relatively noise reduced target. Sentinel-2 relfectance bands, spectral indices, GLASS spectral and radiation products, VIIRS radiation data and annual crop type rasters were acquired and matched as closely as possible to the acquisition date of each SIF observation. All data were aligned to a common 20m grid through the process of area averaging, bilinear interpolation. The irregular shaped OCO-2 polygons were retained as fractional masks in the various input spatial supports (image input dimension) so that the predictions could be averaged over their actual spatial support. To understand the effect of model architecture and input data representation, various models like SIF footprint mean aggregated Random Forest and XGBoost models, area-to-point MLP, single and multi footprint convolutional models, and a U-Net trained on fixed 4km spatial aggregates were trained. These experiments tested whether a spatial model could produce context aware (neighbourhood relationships) fine grid maps and whether OCO-2 aggregation could provide a more stable training target. The resulting 20m SIF outputs should be treated as model derived downscaled estimates rather than direct SIF validated observations (20m), at most it can be validated at 1km^2 resolution. While the design allowed to inspect the effects of target construction, predictor representation, spatial support and model family within one workflow, it cannot guarantee a controlled comparison of model architecture alone and should therefore be seen as comparisons of complete modelling configurations.

Training, validation and testing partitioning was done in such a way that complete dates, footprints, connected date-product components remained together depending on the requirements of each experiment. The best model averaged at least five same date and same mode OCO-2 observations within each fixed 4km grid cell and trained a U-Net whose output was aggregated through the corresponding footprint weight map a shown in FILLREF (model). Internal evaluation used held out aggregate cells while external evaluation was done by applying the saved model and validation derived calibration to native OCO-2 footprints in two previously unseen MGRS tiles. Model performance was additionally examined by month, year, measurement mode, plant-hardiness zone, SIF to Sentinel-2 temporal alignment and footprint count to identify conditions associated with unstable predictions. For the second stage, the selected model was applied to winter wheat pixels in five Bavarian MGRS tiles and mean aggregated from March to July for each NUTS 3 region and year. Random Forest yield models: crop pure enhanced SIF, raw footprints containing at least 5% winter wheat and all accepted raw footprint model configurations were trained on 2019-2022 year splits and tested on 2024 split. Lastly paired bootstrap resampling of the 48 test regions was then used to quantify uncertainty in the differences between the crop-pure and raw-SIF models.

The best model Spatial aggregate (4km) achieved the strongest internal result of RMSE 0.0701 and an R^2 0.761. Aggregation reduced both target and residual variability, confirming that averaging several OCO-2 observations can provide more stable supervision. But it also narrowed the SIF range and removed most extreme values and thus a larger geographic scope is needed to acquire more extreme samples for the model to understand the conditions resulting in very high SIF values. When the model was transferred to individual footprints (1km^2) in two unseen MGRS tiles, performance decreased to an RMSE of 0.1434 and an R^2 of 0.349. It showed that aggregate accuracy overstates model performance at noisier native support. Regarding model architecture, tree based models produced broadly similar aggregate metrics indicating that convolutional architecture alone does not guarantee much greater predictive accuracy than carefully constructed tabular predictors. The convolutional models neverthless have an edge when it comes to spatial representation because their 20m SIF maps used neighbouring pixel information and preserved more accurate field boundaries and landscape context. For the yield experiment, crop pure enhanced SIF produced the lowest 2024 RMSE at 0.9406 t/ha, compared with 1.0285 t/ha for the 5% wheat raw footprint model and 1.0874 t/ha when all raw footprints were used. This corresponded to RMSE reductions of approximately 8.5% and 13.5%, which is in line with the general literature stating that more isolated SIF information can provide useful crop yield and productivity information [PGZ+20, SGG+18, QLH+22]. The yield experiment should be interpreted primarily as a comparison of SIF isolation strategy rather than as an assessment of overall yield prediction accuracy given the R^2 and 95% bootstrap interval figures 6.7. Compared to the existing highest resolution enhanced SIF products at 500m resolution [GTK+22, DXH+25], the thesis best model's external validation at 1km^2 RMSE $= 0.1434$ and an $R^2 = 0.349$ is comparable to the performance metrics of SIFnet OCO-2 validated result of RMSE $= 0.19$, $R^2 = 0.57$, CNSIF's tower based (nine ChinaSpec sites) SIF validation RMSE $= 0.152$, $R^2 = 0.581$. Though it should be noted exact comparison is not reliable as the existing papers tested on different geographic settings (CONUS, China) and at different spatial scales (Nation wide for SIFnet, small local tower based setting for CNSIF). The yield comparison cannot be done directly with existing papers because of the vast differences in crop type, spatial and temporal setting and should be used more of a general expected range for results, useful predictors and model architecture. The results also agree with SIFnet and CNSIF [GTK+22, DXH+25] in showing the value of spatial predictor structure for mixed land cover and also demonstrates that fine output resolution must not be confused with fine scale validation. Finally, the present OCO-2 gap filling and resolution enhancement framework can predict between satellite tracks without a parent measurement unlike the observation preserving TROPOMI methods, but this also increases the importance of the quality of the predictor variables thereby making independent fine resolution validation with future high resolution SIF data like FLEX ESA 300m SIF an important requirement for future work [CLS+25, ZZZ23].

7.2 Dissemination

The main outputs for this thesis are the evaluated SIF enhancement workflows: the associated data preparation and modelling workflow and evidence concerning the usefulness and limitations of crop pure SIF for regional yield prediction. The most immediate potential users are remote-sensing, SIF, and machine-learning researchers who require methods for combining sparse fluorescence observations with spatially complete high resolution predictors. Agricultural researchers and crop monitoring agencies could use the resulting crop specific SIF summaries together with temperature, weather and soil information to investigate crop productivity & yield and plant stress. In addition, spatially detailed SIF estimates could support research on vegetation phenology, and plant responses to drought or heat [QLH+22, SGG+18, PGZ+20]. The thesis is itself the main dissemination output and documents the datasets, their quality, processing decisions, model configurations, validation design and interpretation limits needed to understand and reproduce this work. The thesis study should be treated as a research prototype with no peer-reviewed publication or integration into an operational monitoring service done. Integration into a larger environment would mean connecting the study's data acquisition, preprocessing, model inference to a GIS platform that can support real time inference, data generation and also support external tasks like agricultural monitoring. Such an integration would require a more automated workflow than the current partially manual approach, with sufficient computing and data storage infrastructure invested.

7.3 Problems Encountered

The Spatial Aggregate (4km) uses a fixed grid support to group 5 or more observations in one grid cell, after which they are averaged to a single value. This was done to make the downstream data preparation and data matching steps more computationally efficient. Future work can conduct density based clustering like in [YWC$^+$19] (which is a more accurate representation of spatial grouping) but for high resolution predictors compared to the paper's 0.05 resolution. Spatial aggregation also reduced the observed SIF range, as there were only a few single footprint SIF samples in the $[0.75, 1.5)$ range for the nine MGRS tiles in Germany, an expanded geographic scope is needed to obtain more very high SIF samples and therefore spatial aggregation can cover this range more reliably. As mentioned in [DKP$^+$22], both spatial and temporal aggregation can be done for OCO-2 SIF data to reduce SIF retrieval noise. But to make relatively high spatial resolution aggregates, temporal aggregation cannot work with OCO-2 data as over a period of 8-16 days, individual single date OCO-2 tracks can be separated by 10s of kilometres thereby increasing the spatial resolution of this aggregation. It would work for large scale studies of natural ecosystems or regional and state wide agriculture, but its coarse spatial resolution limits its usefulness for field level SIF reconstruction and analysis. Combining several OCO-2 dates could also mix different weather conditions and short term vegetation responses. Another issue was the temporal mismatch between SIF and Sentinel-2 optical data. OCO-2 SIF is attached to an individual acquisition date, VIIRS PAR is a daily mean, GLASS FAPAR is an eight-day composite, and Sentinel-2 WASP reflectance represents a monthly composite. The predictors were matched as closely as their temporal definitions allowed: PAR by exact date, FAPAR by the eight-day interval containing the SIF date and Sentinel-2 by its monthly product interval. Therefore the Sentinel-2 data represents more of an average monthly reflectance predictor data and thus could impede instantaneous SIF reconstruction. Native OCO-2 validation as seen in Figure **??** can be noisy because the observed SIF may contain short term physiological responses that are not represented in the temporally aggregated Sentinel-2 predictors (monthly composites).

Many of the previous SIF enhancement studies used MODIS predictors because they provided regular global composites at 250m, 500m resolution making data storage and processing much easier than handling 10m, 20m Sentinel-2 predictors. Moving from 500 m to 20 m increases the number of pixels covering the same area by a factor of 625. This therefore limited the spatial and temporal scope for acquiring Sentinel-2 data, eventhough SIF observation data for Germany ranged to about $\tilde{5}$30k observations for years 2019-2024, the limited scope of Sentinel-2 data reduced it to $\tilde{8}$0k single footprint observations. Increasing the geographical scope could also add more samples from the low and high ends of the SIF distribution and could reduce the model's tendency to predict values near the centre of the training range. Another issue was that annual crop type rasters were themselves produced using Random Forest classification and spatial temporal filtering of Sentinel-1 and Sentinel-2 observations. They are therefore model outputs rather than error free original field records, [GHA$^+$25] reported a winter wheat F1-score of 0.82 and overall crop map accuracy of 75.5%. Therefore classification errors from this data can propagate into the SIF model because they were used as crop composition predictors. The V11r OCO-2 SIF data used for this thesis does not include the correction for rotational Raman scattering (helps reduce SIF retrieval bias) described by Sanghavi et al. [SFN$^+$25] which will be available in future V12 data versions. Given the large inference times for generating 20m SIF maps, the yield experiment was conducted on only five MGRS tiles in Bavaria containing 48 NUTS 3 regions generating a record of 245 region-year records. Although the SIF maps were generated at 20 m resolution, the winter wheat pixels were averaged to one monthly value for each NUTS 3 region (limited by scope of yield data). Therefore much of the field scale spatial information was therefore averaged out before yield prediction. Around roughly 290 NUTS 3 regions are available Germany wide (for harvest yield) potentially generating $\tilde{1}$,450 records, it would require extensive computational resources. Although these problems were not fully resolved within this thesis, they can be integrated in the current workflow with the addition of extra compute resources.

7.4 Outlook

In conclusion, future work should focus on obtaining denser and better corrected SIF observations, improved temporal alignment between SIF observations and predictor variables and evaluate the resulting enhanced SIF data at a spatial scale closer to their 20m output resolution. European Space Agency's (ESA) FLEX mission [CTP+17, DMDB+16], which is scheduled for launch in September 2026, will provide multi band SIF observations at approximately 300m resolution with more spatially and temporally complete samples. This 300m dataset can be the target support for future SIF enhancement downscaling methods or provide validation support for past SIF enhancement methods in Table 3.1 including the thesis's method. Instead of using OCO-2 as the target support, TROPOMI SIF data can be used along with Sentinel-2 data for downscaling as it can include parent magnitude support during model training. As noted by [SFN+25], Build 12 of OCO-2/3 Level-2 products will provide a more stable bias corrected version of current OCO-2/3 products and can be used for the current thesis workflow when it is made available. With the addition of improved compute & storage resources, cloud free Sentinel-2 Level 2A rasters closer to each SIF observation date can be used instead of monthly WASP Sentinel-2 composites as they will provide a much better temporal alignment between SIF and Sentinel-2 predictors. For the agricultural application, confidential parcel level yield observations (versus current NUTS 3 / district level) should be requested from the respective authorities, similar to the reference data used by Brandt et al. [BBB+24] as it will greatly increase the size of the training data and give better understanding of the differences between crop pure and mixed SIF signals when done at field scales. Together, these additions could improve the accuracy, spatial detail and agricultural relevance of future enhanced SIF products.

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Annex

```
<?xml version="1.0" encoding="UTF-8"?>
<widget>
         dummy source code
</widget>
```

Listing 1: Sourcecode Listing

```
dummy text
```

Listing 2: SIP request and response packet[?]